Ferromagnetic scar tower under bulk dephasing: notes on the results so far

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Abstract

We study how the ferromagnetic quantum many-body scar tower of a spin-1/2 chain decays under uniform σ^z dephasing of rate γ, comparing the scar state $|-\rangle^{\otimes L}$ with two non-scar product states, the x-Néel and z-Néel states. With exact density-matrix evolution up to $L = 14$ and quantum trajectories up to $L = 20$, the $1/e$ rate of the scar's transverse magnetization becomes L-independent for $\gamma \gtrsim 0.05$, and its thermodynamic value Γ_∞ vanishes as $\gamma \to 0$, roughly as $\gamma^{1/2}$ for $0.02 \leq \gamma \leq 0.2$. The thermal x-Néel signal instead decays at the closed-system thermalization rate ≈ 4.8 for every $\gamma \leq 0.1$. At $\gamma \leq 0.007$ and fixed $L \leq 20$, the scar rate sits at the finite-size cap $\Gamma_e \approx \gamma(L-1)$. A single-kick model, in which every σ^z kick damages the coherence as one flip evolved with H does, reproduces the measured rate at every L and $\gamma \leq 0.05$ to within $\approx 5\%$, without fit parameters. The thermodynamic lifetime at small γ is therefore fixed by how the damage of a single kick grows in time; $L \leq 24$ does not yet decide between ballistic ($\Gamma_\infty \propto \gamma^{1/2}$) and diffusive ($\propto \gamma^{2/3}$) growth.

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1 Model

Hamiltonian. Spin-1/2 ring of L sites (periodic boundaries), $S_j^\alpha = \sigma_j^\alpha/2$, bond singlet projector $R_{j,j+1} = \frac{1}{4} - \mathbf{S}_j \cdot \mathbf{S}_{j+1}$:

$$H = H_I + hS^z_{\rm tot}, \qquad H_I = \sum_j \Big[\sum_\alpha \big(a_\alpha R_{j,j+1}\sigma^\alpha_{j+2} + b_\alpha \sigma^\alpha_{j-1} R_{j,j+1}\big) + \sum_{\alpha\beta} c^{(j)}_{\alpha\beta}\, \sigma^\beta_{j-1} R_{j,j+1} \sigma^\alpha_{j+2} \Big]. \quad (1)$$

Every term of H_I annihilates the ferromagnetic multiplet $|\Phi_n\rangle \propto (S^-_{\rm tot})^n |\uparrow\cdots\uparrow\rangle$, $n = 0, \ldots, L$, so $H|\Phi_n\rangle = h(L/2 - n)|\Phi_n\rangle$: an equally spaced tower of $L + 1$ exact scars.

Coefficients. $h = 1$, $a = b = (1, 1, 1)$, and $c_{xx} = c_{yy} = c_{zz} = 1$ (all other $c_{\alpha\beta} = 0$), each of the three multiplied on every bond j by an independent factor uniform in $[0.5, 1.5]$. The random c removes translation and reflection symmetry; a and b break SU(2), U(1), spin inversion and complex conjugation. One realization per L (seed $1000 + L$), shared by every run. All results in Secs. 6–8 use these coefficients; the earlier results with different coefficients are collected in Appendix A.

Dephasing.

$$\dot{\rho} = \mathcal{L}\rho = -i[H,\rho] + \gamma \sum_j \left(\sigma_j^z \rho\, \sigma_j^z - \rho\right). \tag{2}$$

In the Pauli basis $\rho = 2^{-L}\sum_P r_P P$ the dissipator is diagonal, $\mathcal{D}(P) = -2\gamma\, n_{xy}(P)\, P$, with $n_{xy}(P)$ the number of sites carrying σ^x or σ^y. A single spin's transverse coherence decays as $e^{-2\gamma t}$. The dissipator does not respect the scars.

Methods. Exact evolution of ρ up to $L = 14$, quantum trajectories (diffusive and jump unravellings) up to $L = 20$, single-kick runs up to $L = 24$, and dense diagonalization of H ($L \le 12$) and $\mathcal{L}$ ($L \le 7$). How each is done, in which basis, and how every quantity is computed: Sec. 3.

2 Theory

This section collects the theory behind what is computed: how a Lindbladian evolves states and observables, why the tower is special, what is known about the weak-dissipation limit, what the other noise channels do, and what each measured quantity means. Numerical methods follow in Sec. 3.

2.1 Lindblad dynamics: forward and backward evolution

The generator. A Markovian open system with Hamiltonian H and jump operators L_l obeys

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H,\rho] + \sum_l \Big(L_l \rho L_l^\dagger - \tfrac{1}{2}\{L_l^\dagger L_l, \rho\}\Big), \tag{3}$$

whose solution $\rho(t) = e^{\mathcal{L}t}(\rho_0)$ is a completely positive, trace-preserving semigroup. Here $L_j = \sqrt{\gamma}\,\sigma_j^z$; since $L_j^\dagger L_j = \gamma\mathbb{1}$, Eq. (3) reduces to the dephasing equation of Sec. 1.

Backward (Heisenberg) evolution. With the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^\dagger B)$, the adjoint generator is

$$\mathcal{L}^\dagger(A) = i[H,A] + \sum_l \Big(L_l^\dagger A L_l - \tfrac{1}{2}\{L_l^\dagger L_l, A\}\Big), \tag{4}$$

as follows from the cyclic property of the trace term by term: $\mathrm{Tr}(A\,[H,\rho]) = \mathrm{Tr}([A,H]\,\rho)$ and $\mathrm{Tr}(A\,L\rho L^\dagger) = \mathrm{Tr}(L^\dagger A L\,\rho)$. Hence every expectation value can be computed in two equivalent ways,

$$\langle A\rangle_t = \mathrm{Tr}\left[A\; e^{\mathcal{L}t}(\rho_0)\right] = \mathrm{Tr}\left[e^{\mathcal{L}^\dagger t}(A)\;\rho_0\right], \tag{5}$$

by evolving the state forward (Schrödinger picture) or the observable backward (Heisenberg picture). Trace preservation is $\mathcal{L}^\dagger(\mathbb{1}) = 0$ for every channel; the channel is *unital*, $\mathcal{L}(\mathbb{1}) = 0$, if $\sum_l [L_l, L_l^\dagger] = 0$, which holds for Hermitian jump operators (all dephasings, depolarizing) and fails for decay.

Backward evolution in Pauli coordinates. In the coordinates $r_P = \mathrm{Tr}(\rho P)$ the forward generator is the real matrix G of Eq. (23), and the backward one is its transpose G^{T} (the Pauli strings form an orthogonal real basis of Hermitian operators). The Hamiltonian part of G is antisymmetric (Sec. 3), so $G_H^{\mathsf{T}} = -G_H$: it is the Hamiltonian part for $-H$. The dephasing part is diagonal, so it is its own transpose. Therefore, for the Hermitian channels,

$$e^{\mathcal{L}^\dagger t}\big|_H = e^{\mathcal{L}t}\big|_{H\to -H}, \tag{6}$$

i.e. an observable is evolved backward by running the forward code with the unitary part reversed and the same damping. For decay and thermal noise the feed term is not symmetric and G^{T} differs from G also in the dissipator.

Spectral decomposition. If $\mathcal{L}$ is diagonalizable, with right eigenmatrices $\mathcal{L}(R_k) = \lambda_k R_k$ and left eigenmatrices $\mathcal{L}^\dagger(\tilde{L}_k) = \lambda_k^* \tilde{L}_k$, normalized so that $\mathrm{Tr}(\tilde{L}_j^\dagger R_k) = \delta_{jk}$,

$$\rho(t) = \sum_k e^{\lambda_k t}\ \mathrm{Tr}(\tilde{L}_k^\dagger \rho_0)\, R_k, \qquad \langle A\rangle_t = \sum_k e^{\lambda_k t}\ \mathrm{Tr}(\tilde{L}_k^\dagger \rho_0)\ \mathrm{Tr}(A R_k). \tag{7}$$

Every eigenvalue has $\mathrm{Re}\,\lambda_k \le 0$; $\lambda = 0$ belongs to the steady states; since $\mathcal{L}$ maps Hermitian to Hermitian operators, eigenvalues come in pairs λ, λ^* with $R \leftrightarrow R^\dagger$. A signal therefore decays as a sum of exponentials, and its late-time rate is $-\mathrm{Re}\,\lambda$ of the slowest mode that overlaps both the initial state (left eigenmatrix, backward) and the observable (right eigenmatrix, forward). In the Pauli basis, $G = G_H + G_D$ with G_H antisymmetric and $-2\gamma L \le G_D \le 0$ diagonal, so $-2\gamma L \le \mathrm{Re}\,\lambda \le 0$ (for any eigenvector v, $\mathrm{Re}(v^\dagger G v)/v^\dagger v = v^\dagger G_D v / v^\dagger v$).

Forward evolution of the state, backward evolution of the observable. The semigroup property $e^{\mathcal{L}t} = e^{\mathcal{L}t_1} e^{\mathcal{L}t_2}$ $(t = t_1 + t_2)$ and Eq. (5) applied to the second factor give

$$\langle A\rangle_{t_1+t_2} = \mathrm{Tr}\left[e^{\mathcal{L}^\dagger t_2}(A)\ e^{\mathcal{L}t_1}(\rho_0)\right] = \left\langle e^{\mathcal{L}^\dagger t_2}(A),\, e^{\mathcal{L}t_1}(\rho_0)\right\rangle, \tag{8}$$

(for Hermitian A): the state is evolved forward to t_1, the observable backward (Heisenberg picture) to t_2, and the two meet in an overlap. With $t_1 = t_2 = t/2$ each object has evolved only for half the time. This matters for compressed representations (Sec. 3.13), whose cost is set by the operator entanglement of the evolved object: it grows with time, so two half-time objects can be far cheaper than one full-time one. The two halves also behave differently here. The forward state starts in the tower and stays a product state without noise; the noise moves weight out of the tower, which then thermalizes. The backward-evolved local observable $A = \sigma_j^x$ grows inside a light cone, but its large strings are damped by the dephasing at $2\gamma n_{xy}$ (the backward generator has the same diagonal as the forward one), so the dissipation itself truncates the operator growth: the idea of dissipation-assisted operator evolution. Both are contractions of the same space–time tensor network: forward and backward evolution cut it at an intermediate time, while the influence-matrix method (Sec. 3.13) contracts it along space instead of time.

Where each picture is used. Forward: the exact evolution of ρ, the trajectories, the single-kick runs. Backward: the probes of Appendix A (operator autocorrelations, $\mathrm{Tr}[A\, e^{\mathcal{L}^\dagger t}(B)] = \mathrm{Tr}[e^{\mathcal{L}t}(A)\, B]$ for Hermitian A), the left eigenvectors of the ED, and the adjoint runs of CLIK (Sec. 3.13), whose Krylov vectors are evolved with $\mathcal{L}^\dagger$.

2.2 Scars, thermalization and entanglement

Eigenstate thermalization. In a chaotic (non-integrable) system, expectation values of local operators in eigenstates are smooth functions of energy equal to their thermal values (ETH), and eigenstates in the middle of the spectrum have volume-law entanglement close to that of random states, the Page value $S_{\mathrm{Page}} = L_A \ln 2 - 2^{2L_A - L - 1}$. Local observables of a generic initial state then relax to their thermal values on an L-independent time scale; for the x- and z-Néel states this is the rate $\Gamma_e \approx 4.8$–5 of Sec. 6.

The tower. Quantum many-body scars are eigenstates that violate ETH. Here they form a tower with a restricted spectrum-generating structure: $|\Phi_{n+1}\rangle \propto S_{\mathrm{tot}}^- |\Phi_n\rangle$ and $E_{n+1} - E_n = -h$, because every term of H_I annihilates the ferromagnetic multiplet (each contains a singlet projector $R_{j,j+1}$, and the multiplet is symmetric under every exchange). Any state in their span evolves as $e^{-iHt}\sum_n c_n |\Phi_n\rangle = \sum_n c_n e^{-ih(L/2-n)t}|\Phi_n\rangle$, which returns to itself up to a phase after $2\pi/h$: perfect revivals, and for the spin coherent state $|-\rangle^{\otimes L}$ a rigid precession of all spins (reference 1). Revivals of this kind require scars (Alhambra–Wilming). In the commutant-algebra language (Moudgalya–Motrunich) the tower spans the common kernel of the local annihilators $R_{j,j+1}$, which is why the dressings a, b, c can be chosen freely.

Scar entanglement. For $|\Phi_n\rangle$ and the cut into L_A and $L-L_A$ sites, the Schmidt weights are hypergeometric, $p_\ell = \binom{L_A}{\ell}\binom{L-L_A}{n-\ell}/\binom{L}{n}$ (Sec. 3). For $n = L/2$, $L_A = L/2$ this distribution has variance $\sigma^2 = \frac{nL_A(L-L_A)(L-n)}{L^2(L-1)} \approx L/16$, and the entropy of a discretized Gaussian, $\frac{1}{2}\ln(2\pi e\sigma^2)$, gives

$$S(\Phi_{L/2}) \approx \tfrac{1}{2}\ln\frac{\pi e L}{8}, \tag{9}$$

logarithmic in L (1.27 at $L = 12$, against the exact 1.32), far below the volume law of the bulk (Sec. 10).

What dephasing does to the tower. By permutation symmetry, the projection of σ_j^z onto the tower is the same for every site, so $P_{\text{tower}}\sigma_j^z P_{\text{tower}} = \frac{1}{L}\sum_k P_{\text{tower}}\sigma_k^z P_{\text{tower}} = \frac{2}{L}S^z_{\text{tot}}P_{\text{tower}}$, i.e. $\langle\Phi_a|\sigma_j^z|\Phi_b\rangle = m_a\delta_{ab}$ with $m_a = 1-2a/L$. A kick σ_j^z on $|\Phi_a\rangle$ keeps the weight m_a^2 in the tower and sends $1-m_a^2$ to $k\neq 0$ spin waves, which are not eigenstates of H and spread and thermalize under H_I. For $|-\rangle^{\otimes L}$, the weight kept is $\langle(2S^z_{\text{tot}}/L)^2\rangle = 1/L$. The dephasing is therefore generic: its jump operators are not in the bond algebra of the scars. Jump operators taken from that algebra (e.g. $R_{j,j+1}\sigma^x_{j+2}$) would annihilate the tower and make it a decoherence-free subspace.

2.3 Weak dissipation: the order of limits and the finite-size cap

The question. At fixed L and $\gamma \to 0$, every decay rate is proportional to γ (perturbation theory, below). Mori showed that the limits do not commute: for generic local observables of a chaotic system, taking $L\to\infty$ first, the slowest relevant Liouvillian eigenvalue stays at a finite distance from 0 as $\gamma \to 0^+$; it converges to a Ruelle–Pollicott resonance, the intrinsic relaxation rate of the closed system. The thermal references behave this way: $\Gamma_\infty \approx 4.8$ for every $\gamma \le 0.1$ (x-Néel). The question here is whether the scar revivals are instead *regular*, $\Gamma_\infty \to 0$ as $\gamma\to 0$, and with which power of γ.

First-order perturbation theory at fixed L. For $\gamma\to 0$, the eigenvalues of $\mathcal{L} = \mathcal{L}_0 + \mathcal{D}$ with $\mathcal{L}_0 = -i[H,\cdot]$ are, to first order, the eigenvalues of $\mathcal{D}$ restricted to each degenerate eigenspace of $\mathcal{L}_0$. The tower coherences $|\Phi_a\rangle\langle\Phi_b|$ share the frequency $-ih(b-a)$; restricted to them, with $\langle\Phi_a|\sigma_j^z|\Phi_b\rangle = m_a\delta_{ab}$,

$$P_{\text{tower}}\,\mathcal{D}\big(|\Phi_a\rangle\langle\Phi_b|\big)P_{\text{tower}} = \gamma\sum_j\big(m_a m_b - 1\big)|\Phi_a\rangle\langle\Phi_b| = -\gamma L\,(1-m_a m_b)\,|\Phi_a\rangle\langle\Phi_b|.$$

The transverse magnetization involves the coherences $(a,b) = (n,n+1)$ with weights $\propto |c_n c_{n+1}|\,\langle\Phi_n|S^+_{\text{tot}}|\Phi_{n+1}\rangle = L\binom{L-1}{n}/2^L$, i.e. n binomial with $L-1$ trials. With $u = m_n = 1-2n/L$: $\mathbb{E}[u] = 1/L$, $\operatorname{Var} u = (L-1)/L^2$, and $m_{n+1} = u - 2/L$, so $\mathbb{E}[m_n m_{n+1}] = (L-2)/L^2$ and the mean first-order rate is

$$\Gamma_{\text{PT}} = \gamma L\big(1-\mathbb{E}[m_n m_{n+1}]\big) = \gamma\Big(L-1+\frac{2}{L}\Big). \tag{10}$$

This is the finite-size cap $\Gamma_e/\gamma \to L-1$ of Sec. 7. It is also the saturated single-kick damage: Eq. (49) with $\bar{D}\equiv D_{\text{sat}}$ gives γD_{sat}, and $D_{\text{sat}}(L=12) = 11.0$ against $L-1+2/L = 11.17$. The first-order result holds as long as the damage of a kick has filled the ring before the signal decays: the decay time $1/\Gamma_{\text{PT}} \approx 1/(\gamma L)$ must exceed the saturation time τ_{sat} of $\bar{D}$.

Damage spreading and the thermodynamic rate (heuristic). Suppose the damage of one kick grows as $\bar{D}_\infty(\tau) \simeq A\tau^\beta$ in an infinite system ($\beta = 1$: ballistic spreading of the defect, $\beta = 1/2$: diffusive) and saturates at $D_{\text{sat}} \approx L$ in a ring of L sites, after $\tau_{\text{sat}} \sim L^{1/\beta}$. The single-kick formula $\gamma\int_0^{t_e}\bar{D} = 1$ then gives

$$\Gamma_\infty \propto \gamma^{1/(1+\beta)}: \qquad \beta = 1 \Rightarrow \Gamma_\infty \propto \gamma^{1/2}, \qquad \beta = \tfrac{1}{2} \Rightarrow \Gamma_\infty \propto \gamma^{2/3}. \tag{11}$$

At finite L the cap applies when $\gamma L\,\tau_{\rm sat} \ll 1$, i.e. for $L \ll L^*(\gamma) \sim \gamma^{-\beta/(1+\beta)}$. In the collapse $\Gamma_e/\gamma = \xi f(L/\xi)$ with $\xi = \gamma^{-\nu}$ (Sec. 7) this means $\nu = \beta/(1+\beta)$ and $\Gamma_\infty \propto \gamma^{1-\nu}$: $\nu = 1/2$ for ballistic and $\nu = 1/3$ for diffusive damage. The measured $\nu = 0.33$ (0.24–0.48) and the local exponent ≈ 0.5 of Γ_∞ do not separate the two, which is why the growth of $\bar{D}$ at long τ and large L (step B2, option C2) is the deciding measurement.

2.4 Other single-site noise channels

The channels of Sec. 3 probe whether the scar lifetime depends on the kind of noise. Their effect on free spins ($H_I = 0$) in the precessing frame follows from the per-site damping table: for $h \gg \gamma$ the in-plane Bloch vector rotates fast, so the in-plane damping is averaged over the x and y components.

- σ^z **dephasing:** both in-plane components decay at 2γ: $m_\perp = e^{-2\gamma t}$ (reference 2). Unital; $|\Phi_0\rangle$, $|\Phi_L\rangle$ are dark states (eigenstates of H and of every σ_j^z).
- σ^x **(or σ^y) dephasing:** damps the y (x) component at 2γ and not the other, so the precessing $m_\perp$ decays at the average rate γ; σ^z decays at 2γ. Unital. A kick σ_k^x reflects spin k about the x axis: at phase φ its coherence becomes $c_k = e^{-2i\varphi}$, so $D(0) = 1 - \cos 2\varphi$, 1 on average instead of 2 for σ^z, consistent with the halved free-spin rate.
- **Depolarizing** ($\sqrt{\gamma/2}\,\sigma^\alpha$): every component decays at 2γ; isotropic and unital.
- **Decay** ($\sqrt{\gamma}\,\sigma^-$): in-plane components decay at $\gamma/2$, and $\langle\sigma_j^z\rangle$ relaxes to -1 at rate γ. Not unital. $|\Phi_L\rangle = |\downarrow\cdots\downarrow\rangle$ is annihilated by every σ_j^- and is an eigenstate of H, hence a pure steady state: the bottom of the scar tower is an attractor.
- **Thermal pair** ($\sqrt{\gamma(\bar{n}+1)}\,\sigma^-$, $\sqrt{\gamma\bar{n}}\,\sigma^+$): in-plane decay at $\gamma(2\bar{n}+1)/2$, $\langle\sigma^z\rangle \to -1/(2\bar{n}+1)$ for free spins; with H_I (which breaks the U(1) symmetry about z) the steady state is not known in closed form.

2.5 What the new observables measure

Purity and operator size. The purity $\mathrm{Tr}\,\rho^2 = 2^{-L}\sum_P r_P^2$ is a global quantity. The Hamiltonian part of G is antisymmetric and conserves $\sum_P r_P^2$, while σ^z dephasing damps each r_P at $2\gamma n_{xy}(P)$, so

$$\frac{d}{dt}\ln \mathrm{Tr}\,\rho^2 = \frac{\sum_P 2 r_P \dot{r}_P}{\sum_P r_P^2} = -4\gamma\,\langle n_{xy}\rangle, \qquad \langle n_{xy}\rangle = \frac{\sum_P n_{xy}(P)\, r_P^2}{\sum_P r_P^2}. \tag{12}$$

For $|-\rangle^{\otimes L}$ the weights r_P^2 are uniform on the 2^L strings of $\mathbb{1}$ and σ^x, so the size n is binomial with mean $L/2$ and the purity first decays at $2\gamma L$, extensive like the fidelity. The size distribution $P(n)$ and the support distribution $P(\ell)$ (length of the shortest arc containing the string) show how H moves weight between strings while the dephasing removes it in proportion to n_{xy}: the interplay of operator spreading and dissipation in the density matrix itself.

Operator entanglement of ρ. The operator Schmidt weights of ρ across a cut are the Schmidt weights of ρ viewed as a state in operator space. They set the cost of a matrix-product representation of ρ: an MPS with bond dimension χ at that cut misses exactly the weight $\sum_{\ell>\chi} w_\ell$ (Sec. 3.13), so $S_{\rm op}$ measures how hard $\rho(t)$ is for tensor networks. A product ρ has $S_{\rm op} = 0$, and the noiseless scar state stays a product. Dephasing makes ρ a mixture of product states, which already has $S_{\rm op} > 0$ through classical correlations; H_I then spreads and entangles the kicked components. This is the mechanism behind the failure of truncated TDVP for this problem (Sec. 11).

Entanglement along trajectories. With Hermitian noise, a diffusive trajectory is a pure state evolved by H and by random single-site unitaries. Single-site unitaries cannot create entanglement, and the noiseless scar state never becomes entangled. So the half-chain entropy of a trajectory measures the entanglement that H generates once the noise has pushed the state out of the tower; it is a direct witness of leaving the scar subspace. Its average depends on the unravelling (diffusive and jump trajectories with the same ρ give different values) and is not a function of ρ.

Entanglement after a single flip. σ^z_k applied to the precessing product state gives again a product state; the entanglement of the blocks $[0, \ell)$ then grows from site k outward. Its light cone measures the spreading velocity of the defect independently of the damage D.

3 How the simulations are done

All codes are in `rust/` (one binary per kind of run). The model coefficients come from `rust/src/parameters.rs`; the random c comes from a generator seeded with $1000 \cdot \text{seed} + L$ (seed 1), so every run at a given L uses the same Hamiltonian. No symmetry is used: the random c removes translation, and every calculation works in the full space, 2^L amplitudes for a pure state or 4^L coefficients for a density matrix. (Only the uniform-c runs of the earlier results in Appendix A used translation orbits.)

3.1 Notation

- **Sites** $j = 0, \ldots, L-1$, periodic. Pauli matrices $\sigma^x, \sigma^y, \sigma^z$ and $\sigma^0 = \mathbb{1}$; σ^α_j acts on site j.
- **Basis states.** $|\boldsymbol{s}\rangle = |s_0 s_1 \ldots s_{L-1}\rangle$ with $\boldsymbol{s} = (s_0, \ldots, s_{L-1}) \in \{0,1\}^L$, and $s_j = 0$ for $|\uparrow\rangle$ ($\sigma^z_j = +1$), $s_j = 1$ for $|\downarrow\rangle$. In the code $\boldsymbol{s}$ is the integer $\sum_j s_j 2^j$.
- **Operations on bit strings.** For $\boldsymbol{\mu}, \boldsymbol{s} \in \{0,1\}^L$: $\boldsymbol{\mu} \oplus \boldsymbol{s}$ is the sum modulo 2 site by site; $\boldsymbol{\mu} \cdot \boldsymbol{s} = \sum_j \mu_j s_j$ is the number of sites where both are 1; $|\boldsymbol{\mu}| = \sum_j \mu_j$; $\boldsymbol{e}_j$ is the string with a single 1 on site j.
- **Pure states** $|\psi\rangle = \sum_{\boldsymbol{s}} \psi(\boldsymbol{s})|\boldsymbol{s}\rangle$, stored as the 2^L complex amplitudes $\psi(\boldsymbol{s})$ (the *computational basis*).

3.2 Pauli strings

On one site, $\sigma^x|s\rangle = |1-s\rangle$, $\sigma^z|s\rangle = (-1)^s|s\rangle$ and $\sigma^y = i\sigma^x\sigma^z$ (check: $i\sigma^x\sigma^z = i\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \sigma^y$).

A Pauli string, a product of one of $\sigma^0, \sigma^x, \sigma^y, \sigma^z$ on every site, is therefore fixed by two bit strings, the flip pattern $\boldsymbol{\mu}$ and the phase pattern $\boldsymbol{\nu}$:

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}} = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}} \prod_j \left(\sigma^x_j\right)^{\mu_j} \left(\sigma^z_j\right)^{\nu_j}, \tag{13}$$

with site j carrying σ^0 for $(\mu_j, \nu_j) = (0,0)$, σ^x for $(1,0)$, σ^z for $(0,1)$ and σ^y for $(1,1)$; the factor $i^{\boldsymbol{\mu}\cdot\boldsymbol{\nu}}$ supplies one i for every σ^y. Acting on a basis state, the σ^z factors act first and give $(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}$, then the σ^x factors flip the bits where $\mu_j = 1$:

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}}|\boldsymbol{s}\rangle = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}}\,(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}\,|\boldsymbol{s}\oplus\boldsymbol{\mu}\rangle. \tag{14}$$

Products. Since $(\sigma^x_j)^2 = (\sigma^z_j)^2 = \mathbb{1}$ and $\sigma^z_j\sigma^x_j = -\sigma^x_j\sigma^z_j$, moving the σ^z factors of the first string past the σ^x factors of the second gives

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}} P_{\boldsymbol{\mu}'\boldsymbol{\nu}'} = \omega\, P_{\boldsymbol{\mu}\oplus\boldsymbol{\mu}',\,\boldsymbol{\nu}\oplus\boldsymbol{\nu}'}, \qquad \omega = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}+\boldsymbol{\mu}'\cdot\boldsymbol{\nu}'-(\boldsymbol{\mu}\oplus\boldsymbol{\mu}')\cdot(\boldsymbol{\nu}\oplus\boldsymbol{\nu}')}\,(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{\mu}'} \in \{\pm 1, \pm i\}. \tag{15}$$

Exchanging the two strings changes the sign $(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{\mu}'}$ into $(-1)^{\boldsymbol{\nu}'\cdot\boldsymbol{\mu}}$, so two strings commute if $\boldsymbol{\nu}\cdot\boldsymbol{\mu}' + \boldsymbol{\nu}'\cdot\boldsymbol{\mu}$ is even and anticommute if it is odd. Every string except $\mathbb{1}$ is traceless, so

$$\operatorname{Tr}\big(P\,Q\big) = 2^L\,\delta_{PQ} \tag{16}$$

for any two strings P, Q (they are Hermitian and square to $\mathbb{1}$).

The Hamiltonian as a sum of strings. With $\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{4}\sum_\alpha \sigma_j^\alpha \sigma_{j+1}^\alpha$, $R_{j,j+1} = \frac{1}{4}\big(\mathbb{1} - \sum_\alpha \sigma_j^\alpha \sigma_{j+1}^\alpha\big)$. A term $c\,R_{j,j+1}\,O$ of H_I, with O its dressing (σ_{j+2}^α, σ_{j-1}^β or $\sigma_{j-1}^\beta\sigma_{j+2}^\alpha$, on sites disjoint from $j, j+1$), becomes $\frac{c}{4}\big(O - \sum_\alpha \sigma_j^\alpha\sigma_{j+1}^\alpha O\big)$: four strings with real coefficients. The field gives $hS^z_{\mathrm{tot}} = \frac{h}{2}\sum_j \sigma_j^z$. Collecting equal strings,

$$H = \sum_P h_P\,P, \qquad h_P \in \mathbb{R}, \tag{17}$$

about 600 strings at $L = 20$, with 101 distinct flip patterns $\boldsymbol{\mu}$ (the identity is dropped).

3.3 Pure states: trajectories and single flips

Applying H. From Eq. (14), $\langle \boldsymbol{s}'|P_{\boldsymbol{\mu\nu}}|\boldsymbol{s}\rangle$ is nonzero only for $\boldsymbol{s} = \boldsymbol{s}' \oplus \boldsymbol{\mu}$, so

$$(H\psi)(\boldsymbol{s}') = \sum_{\boldsymbol{\mu}} \psi(\boldsymbol{s}'\oplus\boldsymbol{\mu}) \sum_{\boldsymbol{\nu}:\,h_{\boldsymbol{\mu\nu}}\neq 0} h_{\boldsymbol{\mu\nu}}\, i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}}\,(-1)^{\boldsymbol{\nu}\cdot(\boldsymbol{s}'\oplus\boldsymbol{\mu})}. \tag{18}$$

The strings are grouped by flip pattern, so every amplitude $\psi(\boldsymbol{s}'\oplus\boldsymbol{\mu})$ is read once per group; for $2^L \geq 2^{15}$ the output amplitudes are split over threads.

Time evolution is the Taylor series $e^{-iH\tau}\psi = \sum_{k\geq 0}\frac{(-i\tau)^k}{k!}H^k\psi$, summed until the norm of the last term is below 10^{-17}, in substeps short enough that $1.5\,\|H\|\,\tau_{\mathrm{sub}} \leq 4$ (the norm $\|H\|$ from 40 power iterations).

Initial states. A spin with Bloch vector $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is $\cos\frac{\theta}{2}|\uparrow\rangle + e^{i\phi}\sin\frac{\theta}{2}|\downarrow\rangle$, so a product state has $\psi(\boldsymbol{s}) = \prod_j u_j(s_j)$ with $u_j(0) = \cos\frac{\theta_j}{2}$, $u_j(1) = e^{i\phi_j}\sin\frac{\theta_j}{2}$. For the scar state $|-\rangle^{\otimes L}$ ($\theta = \pi/2$, $\phi = \pi$), $\psi(\boldsymbol{s}) = 2^{-L/2}(-1)^{|\boldsymbol{s}|}$.

Local magnetization. σ_j^z is diagonal: $\langle\sigma_j^z\rangle = \sum_{\boldsymbol{s}}(-1)^{s_j}|\psi(\boldsymbol{s})|^2$. For the transverse components use $\sigma_j^+ = \frac{1}{2}(\sigma_j^x + i\sigma_j^y) = |\uparrow\rangle\langle\downarrow|_j$, which maps $|\boldsymbol{s}\oplus\boldsymbol{e}_j\rangle$ to $|\boldsymbol{s}\rangle$ when $s_j = 0$:

$$\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle = 2\langle\psi|\sigma_j^+|\psi\rangle = 2\sum_{\boldsymbol{s}:\,s_j=0}\psi(\boldsymbol{s})^*\,\psi(\boldsymbol{s}\oplus\boldsymbol{e}_j). \tag{19}$$

(Check: for $|+\rangle$ both amplitudes are $1/\sqrt{2}$ and the sum gives $\langle\sigma^x\rangle = 1$.) Then $m_\alpha = L^{-1}\sum_j\langle\sigma_j^\alpha\rangle$ and $s_\alpha = L^{-1}\sum_j(-1)^j\langle\sigma_j^\alpha\rangle$.

Lindblad equation and its unravellings. With jump operators $L_j = \sqrt{\gamma}\,\sigma_j^z$ the Lindblad equation

$$\dot\rho = -i[H,\rho] + \sum_j\Big(L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j,\rho\}\Big) \tag{20}$$

is the dephasing equation of Sec. 1, because $L_j^\dagger L_j = \gamma\mathbb{1}$. A trajectory is a random pure state $|\psi(t)\rangle$ whose average $\overline{|\psi(t)\rangle\langle\psi(t)|}$ equals $\rho(t)$.

- **Jump unravelling.** Between jumps a trajectory evolves with $H_{\text{eff}} = H - \frac{i}{2}\sum_j L_j^\dagger L_j = H - \frac{i}{2}\gamma L$. The anti-Hermitian part is a multiple of $\mathbb{1}$, so $e^{-iH_{\text{eff}}\tau}|\psi\rangle = e^{-\gamma L\tau/2}e^{-iH\tau}|\psi\rangle$: the norm squared, the probability of no jump in $[0,\tau]$, is $e^{-\gamma L\tau}$ for every state. The waiting time is therefore exponential with rate γL, sampled as $\tau_w = -\ln\xi/(\gamma L)$ with ξ uniform in $(0,1]$. The jump goes to channel j with probability $\|L_j\psi\|^2/\sum_k\|L_k\psi\|^2 = 1/L$, a uniform site, and replaces $\psi(\boldsymbol{s}) \to (-1)^{s_j}\psi(\boldsymbol{s})$ (a π rotation of spin j about z; the norm is unchanged). Since none of these probabilities depends on the state, the sampling is exact, and the only numerical error is that of $e^{-iH\tau}$. From the scar state, $e^{-iHt}|-\rangle^{\otimes L}$ is the product state of reference 1 (spins along $(-\cos ht, -\sin ht, 0)$, up to a global phase), so it is written down at the time of the first jump instead of being evolved.

- **Diffusive unravelling.** One step of length dt is

$$\psi \;\to\; e^{-iH\,dt/2}\prod_j e^{-i\theta_j\sigma_j^z}\,e^{-iH\,dt/2}\,\psi, \qquad \theta_j = \sqrt{\gamma}\,\Delta W_j,\ \Delta W_j \sim \mathcal{N}(0,dt) \text{ independent.}$$

  For one site, $e^{-i\theta\sigma^z}\rho\,e^{i\theta\sigma^z}$ multiplies the coherence $\langle\uparrow|\rho|\downarrow\rangle$ by $e^{-2i\theta}$ and leaves the diagonal alone. The Gaussian average is $\overline{e^{-2i\theta}} = e^{-2\langle\theta^2\rangle} = e^{-2\gamma\,dt}$, exactly the factor by which $\gamma(\sigma^z\rho\sigma^z - \rho)$ damps that coherence in a time dt. So the noise step reproduces the dephasing exactly, and the only error is the symmetric splitting of H and the noise, $O(dt^2)$ on the average. $dt = \min(0.1, 0.05/\gamma)$.

Trajectory averages: $\bar m_x$, $\bar m_y$ are averaged over the n trajectories first, and $m_\perp = (\bar m_x^2 + \bar m_y^2)^{1/2}$ is formed from the averages (the mean of the per-trajectory $m_\perp$ would not decay to zero). The standard error of $\bar m_\alpha$ is the spread of one trajectory over $\sqrt{n}$; runs of one (L,γ) with different noise seeds are pooled on their common time grid, weighted by their trajectory counts. Validation: both unravellings agree with the exact ρ at $L = 8$, $\gamma = 0.1$ and 0.01 (2000 trajectories, rms deviation ≈ 1 standard error); at the production sizes see Table 4. Production: diffusive, 224 trajectories (112 at $L = 20$), $L = 12, \ldots, 20$; jump, 448 trajectories (224 at $L = 18, 20$, $\gamma < 0.002$), output step 0.25.

Single flip. The noiseless scar state at time $t' = \varphi_m/h$, $\varphi_m = 2\pi m/16$, gets one σ_k^z and is evolved with H alone. On site j the reference-1 value of $\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle$ at time t is $-e^{iht}$, so

$$c_j(\tau) = \frac{\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle}{-e^{ih(t'+\tau)}}, \qquad D(\tau) = \sum_j \big[1 - \operatorname{Re} c_j(\tau)\big]. \tag{21}$$

Without the kick every $c_j = 1$ and $D = 0$; right after it $c_k = -1$, so $D(0) = 2$. All $16 \times L$ runs advance together.

3.4 Density matrices: exact evolution in the Pauli basis

Coordinates. By the orthogonality of the strings, every operator can be expanded as

$$\rho = 2^{-L}\sum_P r_P\,P, \qquad r_P = \operatorname{Tr}(\rho P), \tag{22}$$

and r_P is real for Hermitian ρ. There are 4^L coordinates; $r_{\mathbb{1}} = \operatorname{Tr}\rho = 1$.

Equation of motion. $\dot r_P = \operatorname{Tr}(P\dot\rho)$. For the Hamiltonian part, by the cyclic property,

$$\operatorname{Tr}\big(P\,(-i)[H,\rho]\big) = -i\operatorname{Tr}\big(\rho\,[P,H]\big) = \sum_Q h_Q\,\operatorname{Tr}\big(\rho\,i[Q,P]\big).$$

If Q and P commute the term vanishes. If they anticommute, $i[Q,P] = 2i\,QP$, and $QP = \omega P'$ with P' the string of the product rule and $\omega \in \{\pm 1, \pm i\}$; the product of two anticommuting Hermitian matrices is anti-Hermitian, so $\omega = \pm i$ and $i[Q,P] = \kappa_{QP}\,P'$ with $\kappa_{QP} = 2i\omega = \mp 2$, real. For the dephasing, σ_j^z anticommutes with σ_j^x and σ_j^y and commutes with σ_j^0, σ_j^z, so $\sigma_j^z P \sigma_j^z = -P$ if $\mu_j = 1$ and $+P$ otherwise, and

$$\mathcal{D}(P) = \gamma \sum_j \left(\sigma_j^z P \sigma_j^z - P\right) = -2\gamma\,|\boldsymbol{\mu}|\,P,$$

with $|\boldsymbol{\mu}| = n_{xy}(P)$ the number of sites carrying σ^x or σ^y. $\mathcal{D}$ is self-adjoint for $\mathrm{Tr}(A^\dagger B)$, so $\mathrm{Tr}(P\,\mathcal{D}(\rho)) = \mathrm{Tr}(\mathcal{D}(P)\,\rho) = -2\gamma\,n_{xy}(P)\,r_P$. Together,

$$\dot r_P = \sum_{Q \text{ anticommuting with } P} h_Q\,\kappa_{QP}\,r_{P'(Q,P)} \;-\; 2\gamma\,n_{xy}(P)\,r_P \;\equiv\; \sum_{P''} G_{PP''}\,r_{P''}. \tag{23}$$

G is a real $4^L \times 4^L$ matrix. Its Hamiltonian part is antisymmetric: the same Q couples P' back to P, with $i[Q,P'] = \kappa_{QP'}P$, and inserting $P' = \kappa_{QP}^{-1}\,i[Q,P] = 2i\kappa_{QP}^{-1}QP$ gives $i[Q,P'] = 2iQP' = -4\kappa_{QP}^{-1}P$, i.e. $\kappa_{QP'} = -\kappa_{QP}$ (as $\kappa^2 = 4$). Its dephasing part is diagonal and ≤ 0. So every eigenvalue has $-2\gamma L \le \mathrm{Re}\,\lambda \le 0$.

Initial state. For $\rho = \bigotimes_j \frac{1}{2}(\mathbb{1} + \mathbf{n}_j \cdot \boldsymbol{\sigma}_j)$, $\mathrm{Tr}\left[\frac{1}{2}(\mathbb{1} + \mathbf{n}\cdot\boldsymbol{\sigma})\,\sigma^\alpha\right] = n^\alpha$ for $\alpha = x, y, z$ and 1 for σ^0, so $r_P = \prod_j n_j^{\alpha_j}$ with α_j the Pauli matrix of P on site j and $n^0 \equiv 1$. The scar state ($\mathbf{n}_j = (-1,0,0)$) has $r_P = (-1)^{|\boldsymbol{\mu}|}$ for the strings made of σ^0 and σ^x only ($\boldsymbol{\nu} = \mathbf{0}$), and 0 for all others.

Time stepping. $r(t+dt) = e^{G\,dt} r(t)$, summed as a Taylor series to round-off in substeps with $\|G\,dt_{\mathrm{sub}}\|_\infty \le 4$ (largest absolute row sum). G is stored as a sparse matrix up to $L = 10$; from $L = 11$ the row of P is generated on the fly from the strings Q of H that anticommute with P (matrix-free). At $L = 14$, r has $4^{14} = 2.7 \times 10^8$ entries (14.7 GB in all).

Observables are coordinates: $\langle \sigma_j^\alpha \rangle = \mathrm{Tr}(\rho\,\sigma_j^\alpha) = r_{\sigma_j^\alpha}$. The tower weight is $W = \mathrm{Tr}(P_{\mathrm{tower}}\rho) = 2^{-L}\sum_P r_P\,\mathrm{Tr}(P_{\mathrm{tower}}P)$ with $\mathrm{Tr}(P_{\mathrm{tower}}P) = \sum_n \langle\Phi_n|P|\Phi_n\rangle$ evaluated with Eq. (14). In `probes`, an operator X is evolved in the same coordinates, $x_P = \mathrm{Tr}(XP)/2^L$, and autocorrelations use $\mathrm{Tr}[A\,e^{\mathcal{L}^\dagger t}(B)] = \mathrm{Tr}[e^{\mathcal{L}t}(A)\,B] = 2^L \sum_P a_P(t)\,b_P$ for Hermitian A.

3.5 Spectra and eigenstates

Eigenstates of H (`entanglement`). By Eq. (14), $\langle \boldsymbol{s}'|H|\boldsymbol{s}\rangle = \sum_{\boldsymbol{\mu\nu}} h_{\boldsymbol{\mu\nu}}\,i^{\boldsymbol{\mu}\cdot\boldsymbol{\nu}}(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}\,\delta_{\boldsymbol{s}',\,\boldsymbol{s}\oplus\boldsymbol{\mu}}$. This dense complex $2^L \times 2^L$ matrix is diagonalized with `zheevd` (LAPACK from Intel MKL), $L \le 12$. The scars are the Dicke states

$$|\Phi_n\rangle = \binom{L}{n}^{-1/2} \sum_{\boldsymbol{s}:\,|\boldsymbol{s}|=n} |\boldsymbol{s}\rangle, \qquad n = 0, \ldots, L, \tag{24}$$

(n down spins), and the tower weight of an eigenstate $|E\rangle$ is $\sum_n |\langle\Phi_n|E\rangle|^2$; $P_{\mathrm{tower}} = \sum_n |\Phi_n\rangle\langle\Phi_n|$.

Liouvillian modes (`ed`, `lindblad_entanglement`). The matrix G of Eq. (23) is assembled densely and diagonalized with `dgeev`: $Gr = \lambda r$. Since $\mathcal{L}$ acts on coordinates as G, the operator $X = \sum_P r_P\,P$ (a $2^L \times 2^L$ matrix, built with Eq. (14)) is a right eigenmatrix, $\mathcal{L}(X) = \lambda X$. Its tower weight is $\|P_{\mathrm{tower}} X P_{\mathrm{tower}}\|/\|X\|$ (Frobenius norms). Because G is real, r^* is an eigenvector with eigenvalue λ^*, and since every P is Hermitian, $X(\lambda^*) = X(\lambda)^\dagger$. Degenerate eigenvalues ($|\Delta\lambda| < 10^{-8}$) have no unique eigenmatrix; there the basis is the one that diagonalizes $\mathrm{Tr}(X_i^\dagger H X_l)$ against $\mathrm{Tr}(X_i^\dagger X_l)$ within the eigenspace (generalized eigenproblem, `zhegv`). This splits the edge pairs $|\Phi_0\rangle\langle\Phi_n|$, $|\Phi_{L-n}\rangle\langle\Phi_L|$ exactly and, at $\gamma = 0$, returns single $|n\rangle\langle m|$ of eigenstates of H. The dense G limits this to $L \le 7$ (2.1 GB at $L = 7$).

3.6 Entanglement entropy

The cut is $A =$ sites $0, \dots, L_A - 1$, $L_A = \lfloor L/2 \rfloor$, and B the rest; on the ring A has two boundaries. Split $\boldsymbol{s} = (\boldsymbol{s}_A, \boldsymbol{s}_B)$ and write the amplitudes as a $2^{L_A} \times 2^{L-L_A}$ matrix $\Psi_{\boldsymbol{s}_A \boldsymbol{s}_B} = \psi(\boldsymbol{s})$, so that $|\psi\rangle = \sum_{\boldsymbol{s}_A, \boldsymbol{s}_B} \Psi_{\boldsymbol{s}_A \boldsymbol{s}_B} |\boldsymbol{s}_A\rangle \otimes |\boldsymbol{s}_B\rangle$. Tracing out B,

$$\rho_A = \mathrm{Tr}_B |\psi\rangle\langle\psi| = \sum_{\boldsymbol{s}_A, \boldsymbol{s}'_A} \Big(\sum_{\boldsymbol{s}_B} \Psi_{\boldsymbol{s}_A \boldsymbol{s}_B} \Psi^*_{\boldsymbol{s}'_A \boldsymbol{s}_B} \Big) |\boldsymbol{s}_A\rangle\langle \boldsymbol{s}'_A| = \Psi \Psi^\dagger. \tag{25}$$

Its eigenvalues p_ℓ (`zheevd`) are the squared singular values of Ψ, i.e. the squared Schmidt coefficients in $|\psi\rangle = \sum_\ell \sqrt{p_\ell}\, |\varphi_\ell\rangle_A |\chi_\ell\rangle_B$ (orthonormal $|\varphi_\ell\rangle_A$, $|\chi_\ell\rangle_B$), and

$$S_A = -\sum_\ell p_\ell \ln p_\ell. \tag{26}$$

Reference values. A random state gives on average the Page value $S_{\mathrm{Page}} = L_A \ln 2 - 2^{2L_A - L - 1}$ (for $L_A \le L - L_A$). For a Dicke state, the strings with n down spins split into ℓ down spins in A and $n - \ell$ in B, so $|\Phi_n\rangle = \sum_\ell \sqrt{p_\ell}\, |\Phi^{(A)}_\ell\rangle |\Phi^{(B)}_{n-\ell}\rangle$ with Dicke states of the two halves and

$$p_\ell = \binom{L_A}{\ell} \binom{L - L_A}{n - \ell} \Big/ \binom{L}{n}. \tag{27}$$

3.7 Operator entanglement

An operator ρ on the chain is a vector in the space of operators, with the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^\dagger B)$, and it can be Schmidt-decomposed across the same cut. Expand it in the product basis $|\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A| \otimes |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|$,

$$\rho = \sum M_{(\boldsymbol{s}_A \boldsymbol{s}'_A), (\boldsymbol{s}_B \boldsymbol{s}'_B)}\, |\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A| \otimes |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|, \qquad M_{(\boldsymbol{s}_A \boldsymbol{s}'_A), (\boldsymbol{s}_B \boldsymbol{s}'_B)} = \langle \boldsymbol{s}_A \boldsymbol{s}_B | \rho | \boldsymbol{s}'_A \boldsymbol{s}'_B \rangle, \tag{28}$$

a $4^{L_A} \times 4^{L-L_A}$ matrix (the "realigned" ρ). Its singular value decomposition $M = \sum_\ell s_\ell\, u_\ell w_\ell^\dagger$ gives

$$\rho = \sum_\ell s_\ell\, A_\ell \otimes B_\ell, \qquad A_\ell = \sum_{\boldsymbol{s}_A \boldsymbol{s}'_A} (u_\ell)_{\boldsymbol{s}_A \boldsymbol{s}'_A} |\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A|, \quad B_\ell = \sum_{\boldsymbol{s}_B \boldsymbol{s}'_B} (w^*_\ell)_{\boldsymbol{s}_B \boldsymbol{s}'_B} |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|, \tag{29}$$

with $\mathrm{Tr}(A^\dagger_\ell A_{\ell'}) = \mathrm{Tr}(B^\dagger_\ell B_{\ell'}) = \delta_{\ell\ell'}$ and $\sum_\ell s_\ell^2 = \mathrm{Tr}(\rho^\dagger \rho)$. With the normalized weights $w_\ell = s_\ell^2 / \sum_{\ell'} s_{\ell'}^2$,

$$S_{\mathrm{op}}(\rho) = -\sum_\ell w_\ell \ln w_\ell, \qquad 0 \le S_{\mathrm{op}} \le 2 L_A \ln 2; \tag{30}$$

the s_ℓ^2 are computed as the eigenvalues of $M M^\dagger$ ($4^{L_A} \times 4^{L_A}$, `zheevd`).

Pure states. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \sum_\ell \sqrt{p_\ell} |\varphi_\ell\rangle_A |\chi_\ell\rangle_B$,

$$\rho = \sum_{\ell\ell'} \sqrt{p_\ell p_{\ell'}}\, |\varphi_\ell\rangle\langle\varphi_{\ell'}| \otimes |\chi_\ell\rangle\langle\chi_{\ell'}|.$$

The operators $|\varphi_\ell\rangle\langle\varphi_{\ell'}|$ are orthonormal for $\mathrm{Tr}(A^\dagger B)$, and so are the $|\chi_\ell\rangle\langle\chi_{\ell'}|$, so this is already the operator Schmidt decomposition, with one term per pair (ℓ, ℓ') and weights $w_{\ell\ell'} = p_\ell p_{\ell'}$ (they sum to 1). Hence $S_{\mathrm{op}} = -\sum_{\ell\ell'} p_\ell p_{\ell'} \ln(p_\ell p_{\ell'}) = -2 \sum_\ell p_\ell \ln p_\ell = 2 S_A$. In the same way, for two eigenstates with Schmidt decompositions $|n\rangle = \sum_\ell \sqrt{p_\ell} |\varphi_\ell\rangle_A |\chi_\ell\rangle_B$ and $|m\rangle = \sum_{\ell'} \sqrt{q_{\ell'}} |\varphi'_{\ell'}\rangle_A |\chi'_{\ell'}\rangle_B$ (different bases),

$$|n\rangle\langle m| = \sum_{\ell\ell'} \sqrt{p_\ell q_{\ell'}}\, |\varphi_\ell\rangle\langle\varphi'_{\ell'}| \otimes |\chi_\ell\rangle\langle\chi'_{\ell'}|$$

is again a Schmidt decomposition (the $|\varphi_\ell\rangle\langle\varphi'_{\ell'}|$ are orthonormal), with weights $p_\ell q_{\ell'}$, and

$$S_{\mathrm{op}}\big(|n\rangle\langle m|\big) = S_n + S_m \tag{31}$$

exactly (checked by SVD for random states, 2×10^{-15}). Fig. 16 is computed this way from the eigenstate entropies, without further diagonalization.

Liouvillian modes. An eigenmatrix X is not Hermitian, and its normalization and phase are arbitrary. The two positive, unit-trace operators

$$\rho_L = \frac{XX^\dagger}{\mathrm{Tr}(XX^\dagger)}, \qquad \rho_R = \frac{X^\dagger X}{\mathrm{Tr}(X^\dagger X)} \tag{32}$$

are density matrices; we use $S_L = S_{\mathrm{op}}(\rho_L)$, $S_R = S_{\mathrm{op}}(\rho_R)$ and the energies $E_{L,R} = \mathrm{Tr}(\rho_{L,R} H)$. For $X = |n\rangle\langle m|$, $\rho_L = |n\rangle\langle n|$ and $S_L = 2S_n$. Since $X(\lambda^*) = X(\lambda)^\dagger$, $\rho_L(\lambda^*) = \rho_R(\lambda)$, and the (E_R, S_R) set equals the (E_L, S_L) set.

3.8 Other single-site noise channels

Besides the σ^z dephasing of all the main results, the code (`rust/src/noise.rs`) has five more uniform single-site channels, each a set of jump operators on every site j:

channel	jump operators	
z (default)	$\sqrt{\gamma}\,\sigma_j^z$	dephasing
x, y	$\sqrt{\gamma}\,\sigma_j^x$, $\sqrt{\gamma}\,\sigma_j^y$	dephasing along x or y
depolarizing	$\sqrt{\gamma/2}\,\sigma_j^\alpha$, $\alpha = x, y, z$	every Pauli factor decays at 2γ
decay	$\sqrt{\gamma}\,\sigma_j^-$, $\sigma^- = \|\downarrow\rangle\langle\uparrow\|$	amplitude damping towards $\|\downarrow\rangle$
thermal	$\sqrt{\gamma(\bar{n}+1)}\,\sigma_j^-$ and $\sqrt{\gamma\bar{n}}\,\sigma_j^+$	relaxation to a thermal spin

The depolarizing normalization makes every single-site Pauli factor decay at 2γ, the rate of the transverse coherence under σ^z dephasing.

Coordinates. For jump operators L_k the dissipator is $\mathcal{D}(\rho) = \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$, and by the cyclic property

$$\dot{r}_P\big|_{\mathcal{D}} = \mathrm{Tr}\left(P\,\mathcal{D}(\rho)\right) = \mathrm{Tr}\left(\mathcal{D}^\dagger(P)\,\rho\right), \qquad \mathcal{D}^\dagger(P) = \sum_k \left(L_k^\dagger P L_k - \tfrac{1}{2}\{L_k^\dagger L_k, P\}\right). \tag{33}$$

Every L_k acts on one site, so $\mathcal{D}^\dagger$ is a sum over sites of maps acting on the factor of P on that site only, and it suffices to know $\mathcal{D}^\dagger$ on $\sigma^0, \sigma^x, \sigma^y, \sigma^z$.

Hermitian channels. For $L = \sqrt{g}\,\sigma^\alpha$, $L^\dagger L = g\,\mathbb{1}$ and $\mathcal{D}^\dagger(P) = g(\sigma^\alpha P \sigma^\alpha - P)$, which is 0 if the factor P commutes with σ^α and $-2gP$ if it anticommutes. For σ^z dephasing ($g = \gamma$) this is the familiar -2γ on σ^x and σ^y. For the depolarizing channel ($g = \gamma/2$ for each α) a non-identity factor anticommutes with two of the three σ^α, so it decays at $2 \cdot 2 \cdot \gamma/2 = 2\gamma$.

Decay. With $L = \sqrt{\gamma}\,\sigma^-$, $\sigma^- = |\downarrow\rangle\langle\uparrow|$, $L^\dagger = \sqrt{\gamma}\,|\uparrow\rangle\langle\downarrow|$ and $L^\dagger L = \gamma|\uparrow\rangle\langle\uparrow| = \frac{\gamma}{2}(\mathbb{1} + \sigma^z)$:

- $\mathcal{D}^\dagger(\mathbb{1}) = \gamma|\uparrow\rangle\langle\uparrow| - \gamma|\uparrow\rangle\langle\uparrow| = 0$ (trace preservation);
- $\mathcal{D}^\dagger(\sigma^z) = \gamma\,|\uparrow\rangle\langle\downarrow|\,\sigma^z|\downarrow\rangle\langle\uparrow| - \frac{\gamma}{2}\{|\uparrow\rangle\langle\uparrow|, \sigma^z\} = -\gamma|\uparrow\rangle\langle\uparrow| - \gamma|\uparrow\rangle\langle\uparrow| = -\gamma(\mathbb{1} + \sigma^z)$;
- $\mathcal{D}^\dagger(\sigma^x) = \gamma\,|\uparrow\rangle\langle\downarrow|\,\sigma^x|\downarrow\rangle\langle\uparrow| - \frac{\gamma}{2}\{|\uparrow\rangle\langle\uparrow|, \sigma^x\} = 0 - \frac{\gamma}{2}\sigma^x$, since $\langle\downarrow|\,\sigma^x\,|\downarrow\rangle = 0$ and $|\uparrow\rangle\langle\uparrow|\sigma^x + \sigma^x|\uparrow\rangle\langle\uparrow| = \sigma^x$; the same for σ^y.

The thermal channel adds $\sqrt{\gamma\bar{n}}\,\sigma^+$, which by exchanging $\uparrow \leftrightarrow \downarrow$ gives $\mathcal{D}^\dagger(\sigma^z) = \gamma\bar{n}(\mathbb{1} - \sigma^z)$ and $-\frac{\gamma\bar{n}}{2}$ on $\sigma^{x,y}$; with $\gamma(\bar{n}+1)$ for the decay part the sum is $\mathcal{D}^\dagger(\sigma^z) = -\gamma(2\bar{n}+1)\sigma^z - \gamma\mathbb{1}$.

Summary per unit of γ ($\mathcal{D}^\dagger$ on the single-site factors):

channel	$\mathbb{1}$	σ^x	σ^y	σ^z
z	0	$-2\sigma^x$	$-2\sigma^y$	0
x	0	0	$-2\sigma^y$	$-2\sigma^z$
y	0	$-2\sigma^x$	0	$-2\sigma^z$
depolarizing	0	$-2\sigma^x$	$-2\sigma^y$	$-2\sigma^z$
decay	0	$-\frac{1}{2}\sigma^x$	$-\frac{1}{2}\sigma^y$	$-\sigma^z-\mathbb{1}$
thermal	0	$-\frac{2\bar{n}+1}{2}\sigma^x$	$-\frac{2\bar{n}+1}{2}\sigma^y$	$-(2\bar{n}+1)\sigma^z-\mathbb{1}$

Summing over the sites, a string $P=P_{\boldsymbol{\mu\nu}}$ is damped at the sum of the diagonal entries of its factors. For decay and thermal there is in addition a *feed*: for every site j where P carries σ^z ($\mu_j=0$, $\nu_j=1$), the term $-\gamma\mathbb{1}$ of $\mathcal{D}^\dagger(\sigma^z_j)$ gives

$$\dot r_P\big|_{\text{feed}} = -\gamma \sum_{j:\,\mu_j=0,\,\nu_j=1} r_{P^{(j)}}, \qquad P^{(j)}=P_{\boldsymbol{\mu},\,\boldsymbol{\nu}\oplus\boldsymbol{e}_j} \tag{34}$$

(P with that σ^z replaced by $\mathbb{1}$). This off-diagonal term is implemented only in the full Pauli basis. On one site, $\dot r_z=-\gamma(2\bar n+1)r_z-\gamma r_{\mathbb{1}}$ with $r_{\mathbb{1}}=1$, so $\langle\sigma^z\rangle\to-1/(2\bar n+1)$, i.e. -1 for the decay ($\bar n=0$). The decay drives every spin to $|{\downarrow}\rangle$, i.e. the chain to $|\Phi_L\rangle=|{\downarrow}\cdots{\downarrow}\rangle$, a state of the tower.

Diffusive unravelling of the Hermitian channels. As for σ^z, each Hermitian jump operator $\sqrt{g}\,\sigma^\alpha_j$ becomes a random rotation $e^{-i\theta\sigma^\alpha_j}$ with $\theta\sim\mathcal{N}(0,g\,dt)$ in every step. In the eigenbasis of σ^α the rotation multiplies the coherence by $e^{-2i\theta}$, whose average $e^{-2g\,dt}$ is exactly the damping by the channel of $\sqrt{g}\,\sigma^\alpha$ over dt. Explicitly, on the pair $(\psi(\boldsymbol{s}),\psi(\boldsymbol{s}\oplus\boldsymbol{e}_j))$ with $s_j=0$: $e^{-i\theta\sigma^x}=\cos\theta-i\sin\theta\,\sigma^x$ and $e^{-i\theta\sigma^y}=\begin{pmatrix}\cos\theta&-\sin\theta\\ \sin\theta&\cos\theta\end{pmatrix}$. The depolarizing channel uses three independent rotations per site, about x, y and z, each with $g=\gamma/2$; their product reproduces the sum of the three generators up to $O(dt^2)$, like the splitting with H. The jump unravelling is written for σ^z only; decay and thermal have non-Hermitian jump operators and are run with the exact ρ evolution only.

3.9 Operator observables of ρ

The exact runs can also write (option `--obs-dt`), from the coordinates r_P:

- the **purity** $\operatorname{Tr}\rho^2=2^{-2L}\sum_{P,Q}r_Pr_Q\operatorname{Tr}(PQ)=2^{-L}\sum_P r_P^2$;
- the distribution of ρ over strings, with weights $w_P=r_P^2/\sum_Q r_Q^2$ (identity included, $w_{\mathbb{1}}=1/\sum_Q r_Q^2$), and from it: the **size** distribution $P(n)=\sum_{P:\,n(P)=n}w_P$, $n(P)=|\boldsymbol{\mu}\vee\boldsymbol{\nu}|$ the number of non-identity sites (sites with $\mu_j=1$ or $\nu_j=1$); the mean $\langle n_{xy}\rangle=\sum_P w_P|\boldsymbol{\mu}|$; the **support** distribution $P(\ell)$, with $\ell(P)$ the length of the shortest arc of the ring that contains all non-identity sites of P (L minus the longest circular run of identity sites; $\ell=0$ for $\mathbb{1}$).

Purity identity for σ^z dephasing. With $\dot r=Gr$, $G=A-\Gamma$ (A antisymmetric from H, Γ diagonal with entries $2\gamma n_{xy}(P)$, Eq. (23)),

$$\frac{d}{dt}\sum_P r_P^2 = 2\,r^{\mathsf T}Gr = -2\,r^{\mathsf T}\Gamma r = -4\gamma\sum_P n_{xy}(P)\,r_P^2 \quad\Rightarrow\quad \frac{d\ln\operatorname{Tr}\rho^2}{dt}=-4\gamma\,\langle n_{xy}\rangle, \tag{35}$$

because $r^{\mathsf T}Ar=0$: the Hamiltonian conserves $\sum_P r_P^2$, and only the dephasing, through the weight on strings with σ^x or σ^y, lowers the purity.

Operator entanglement of ρ (option `--op-ent-dt`). Every string factorizes across the cut, $P=P_A\otimes P_B$, and the normalized strings $P_A/2^{L_A/2}$, $P_B/2^{(L-L_A)/2}$ are orthonormal for $\operatorname{Tr}(X^\dagger Y)$. So

$$\rho=2^{-L/2}\sum_{P_A,P_B}R_{P_AP_B}\,\frac{P_A}{2^{L_A/2}}\otimes\frac{P_B}{2^{(L-L_A)/2}}, \qquad R_{P_AP_B}=r_{P_A\otimes P_B}, \tag{36}$$

is already written in orthonormal product bases, and the operator Schmidt coefficients of ρ are proportional to the singular values of the real $4^{L_A} \times 4^{L-L_A}$ matrix R. The normalized weights $w_\ell = s_\ell^2 / \sum s^2$ come from the eigenvalues of RR^{T} on the smaller side (a 4096×4096 problem at $L = 12$), and give S_{op} and the Rényi-2 entropy $-\ln \sum_\ell w_\ell^2$ as in Sec. 3.

3.10 Entanglement along trajectories and after a single flip

- `states_traj --ent-dt`: for every trajectory, the half-chain S_A and the Rényi-2 entropy $S_2 = -\ln \mathrm{Tr}\, \rho_A^2$ of its pure state, with $\rho_A = \Psi\Psi^\dagger$ as above, averaged over the trajectories (with standard errors). This *diffusive-trajectory entanglement* depends on the unravelling (diffusive and jump trajectories with the same average ρ have different mean S_A), so it is not a property of ρ. The operator entanglement, purity, size and support above are.
- `single_flip --ent-dt`: S and S_2 of the blocks $[0, \ell)$, $\ell = 1, \ldots, L/2$ (`--ent-max`), of every single-flip run: how the entanglement created by one kick spreads.

3.11 Eigenvalues without LAPACK

The entanglement spectra are computed in `rust/src/entanglement.rs` without external libraries, so the same binaries run on every machine. A real symmetric $n \times n$ matrix is brought to tridiagonal form by $n-2$ Householder reflections $H_k = \mathbb{1} - \beta v v^{\mathsf{T}}$ acting on the trailing block, $B \to B - vw^{\mathsf{T}} - wv^{\mathsf{T}}$ with $p = \beta B v$, $w = p - \frac{\beta}{2}(v^{\mathsf{T}} p)\, v$. The column is first divided by its largest entry: a tiny column would otherwise make $v^{\mathsf{T}} v$ underflow and $\beta = 2/v^{\mathsf{T}} v$ overflow (this produced NaN for the scar state, whose amplitudes carry round-off imaginary parts $\approx 4 \times 10^{-17}$). The tridiagonal eigenvalues follow from implicit QL iterations with Wilkinson shifts; an off-diagonal element is set to zero when it is below ε times the sum of its two diagonal neighbours *or* below ε times the largest row sum of the tridiagonal matrix (the absolute test is needed for the many zero eigenvalues of a low-rank ρ_A). A Hermitian $M = X + iY$ is handled through the real symmetric matrix $\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}$, which has every eigenvalue of M twice. Speed on one core ≈ 5 GFLOP/s ($n = 1024$ in 0.3 s, $n = 2048$ in 3.4 s); inside the parallel loops over trajectories or single-flip runs it runs on one thread per matrix.

3.12 Run options and file names

- `--channel z|x|y|dep|decay|thermal` (and `--nbar`); the output files get `_chx`, `_chy`, `_chdep`, `_chdecay` or `_chthermal_n<`$\bar{n}$`>` (nothing for σ^z).
- `--seed` N: another random-c realization (generator seeded with $1000N+L$); for $N \neq 1$ every output file gets `_seed<`N`>` and the realization is `states_realization_L<L>_seed<`N`>.csv`, so the seed-1 files are never overwritten.
- `--stop-te` F: the run also stops at $t = F\, t_1$, t_1 the first time the signal drops below $1/e$. With $F = 1.5$ this is enough for Γ_e without knowing t_e in advance (as for the new channels).
- New outputs:
  - `states_obs_*.csv`: purity, $\langle n \rangle$, $\langle n_{xy} \rangle$, S_{op}, $S_{2,\mathrm{op}}$, $P(n)$, $P(\ell)$;
  - `states_traj_ent_*.csv`: S_A, S_2 and their errors;
  - `single_flip_ent_*.csv`: S, S_2 per block.

Tests of the new code (`cargo test --release`, 17 tests in all):

- every new channel with free spins ($H_I = 0$, random product state) against the single-spin Bloch equations $\dot{r}_x = -h r_y + \gamma d_x r_x$, $\dot{r}_y = h r_x + \gamma d_y r_y$, $\dot{r}_z = \gamma d_z r_z + \gamma f$ (d the diagonal entries, f the feed) integrated by RK4: agreement to 10^{-10};

- diffusive trajectories with σ^x dephasing and with depolarizing noise against the exact ρ with the full H ($L = 4$, 3000 trajectories): within 5σ;
- the purity identity above, by finite differences ($L = 6$);
- eigenvalues of random symmetric (n up to 600, both code paths) and Hermitian matrices against $\operatorname{Tr} M$, $\operatorname{Tr} M^2$, $\operatorname{Tr} M^3$; Dicke-state block entropies against the exact formula for every block size; product states, including the scar state as the trajectories build it, give 0, and the GHZ state $\ln 2$;
- $S_{\mathrm{op}}(\rho) = 2S_A(\psi)$ and $S_{2,\mathrm{op}} = 2S_2$ for a pure entangled state evolved independently as ρ in the Pauli basis and as a state vector ($L = 6$, 10^{-8}).

3.13 Tensor networks: ρ as a matrix product state, TDVP, influence matrix, CLIK

ρ as an MPS. The Pauli coordinates factorize over sites, $r_P = r_{p_0 p_1 \ldots p_{L-1}}$ with $p_j \in \{0, x, y, z\}$ the factor of P on site j, so ρ is a vector in $(\mathbb{R}^4)^{\otimes L}$ and can be written as a matrix product state

$$r_{p_0 \ldots p_{L-1}} = A_0^{p_0} A_1^{p_1} \cdots A_{L-1}^{p_{L-1}}, \tag{37}$$

with $\chi_j \times \chi_{j+1}$ matrices A_j^p ($\chi_0 = \chi_L = 1$; ITensor, site type `PauliVec`, `julia/lindblad_tdvp.jl`). The singular values of the matrix $r_{(p_0 \ldots p_{j-1}),(p_j \ldots p_{L-1})}$ are the operator Schmidt values of ρ across the cut between sites $j-1$ and j (Sec. 3, the strings being orthogonal with equal norms). The representation is exact for $\chi_j = 4^{\min(j, L-j)}$. A truncation to bond dimension χ keeps the χ largest Schmidt values and, in the normalized vector, loses the weight $\varepsilon^2 = \sum_{\ell > \chi} w_\ell$. The operator entanglement S_{op} of $\rho(t)$ therefore sets the cost: $\chi \gtrsim e^{S_{\mathrm{op}}}$ is needed for small errors. Expectation values are contractions: $\langle \sigma_j^\alpha \rangle = r_{0 \ldots \alpha_j \ldots 0}$, read off by projecting every other site on $p = 0$.

The generator as an MPO. Left multiplication by a Pauli matrix permutes Pauli matrices with phases, $\sigma^a \sigma^b = \sum_c (A^a + iB^a)_{cb}\, \sigma^c$ with real 4×4 matrices A^a, B^a (A^a the symmetric part: $\sigma^a \sigma^0 = \sigma^a$, $\sigma^a \sigma^a = \sigma^0$; B^a the antisymmetric part from $\sigma^a \sigma^b = i\epsilon_{abc}\sigma^c$), and right multiplication gives $A^a - iB^a$. For a string P with factors a_j, $[P, X]$ acts on the coordinates as $\bigotimes_j (A^{a_j} + iB^{a_j}) - \bigotimes_j (A^{a_j} - iB^{a_j})$, so

$$-i[P, \cdot\,] \;\longrightarrow\; 2\,\mathrm{Im} \bigotimes_j \left(A^{a_j} + iB^{a_j}\right), \tag{38}$$

a real operator, and the dephasing is the diagonal $-2\gamma \sum_j (\text{projector on } p_j \in \{x, y\})$. Summing over the strings of H gives G as a real matrix product operator; the periodic boundary enters as long-range terms.

TDVP: forward and backward sweeps. The time-dependent variational principle (Dirac–Frenkel) keeps the state on the manifold $\mathcal{M}_\chi$ of MPS of bond dimension χ by projecting the generator onto its tangent space,

$$\frac{d}{dt}|r\rangle = \mathcal{P}_{T_r \mathcal{M}_\chi}\, G|r\rangle. \tag{39}$$

In mixed-canonical gauge the tangent-space projector decomposes into one-site and bond terms, $\mathcal{P}_T = \sum_j \mathcal{P}_j^{(1)} - \sum_j \mathcal{P}_{j,j+1}^{(0)}$, and the projector-splitting integrator solves the terms one after the other. In a left-to-right sweep, the site tensor A_j is evolved *forward* by δt with its local effective generator, split by a QR decomposition into a left-orthonormal tensor and a bond matrix, and the bond matrix is evolved *backward* by δt (the minus sign of $\mathcal{P}^{(0)}$) before it is absorbed into A_{j+1}. A right-to-left sweep with $\delta t/2$ after a left-to-right one with $\delta t/2$ makes the step symmetric and second order. The two-site variant evolves two-site tensors forward and single-site tensors backward, and splits the two-site tensor by an SVD, so the bond dimension can grow up to $\chi_{\max}$;

the truncation is the only approximation beyond the projection. The local exponentials are applied with Krylov (Arnoldi, G is not symmetric) methods. Two points matter here:

- *Projection error at small bonds:* terms of G that act on more than two sites (the range-4 terms of H, the long-range periodic terms) leave the two-site tangent space while the bonds are small, and two-site TDVP then misses them. A global subspace expansion (enlarging the bond basis with Krylov vectors $G^k|r\rangle$ of the full generator before the sweep) removes this; once every bond has its full dimension TDVP is exact.
- *Truncation error:* at fixed χ it is set by the operator Schmidt spectrum of $\rho(t)$, i.e. by S_{op}.

Backward (Heisenberg) evolution of an observable uses the same machinery with G^{T}, i.e. with $H \to -H$ (Sec. 2.1).

Influence matrix and transverse contraction. The standard contraction follows time: ρ (or A) is evolved step by step as an MPS in space, and its cost is set by the operator entanglement in space. The influence-matrix approach contracts the same network along space instead.

- *The space–time network.* With a Trotter step δt, $e^{\mathcal{L}\delta t} \approx \mathcal{D}\,\mathcal{G}$, where $\mathcal{G}$ is a brickwork of local unitary channels $X \mapsto U_g X U_g^\dagger$ (the terms of H grouped into gates on a few neighbouring sites) and $\mathcal{D} = \bigotimes_j e^{\mathcal{D}_j \delta t}$ the single-site dephasing. In the Pauli (vectorized) representation each channel is a real tensor whose legs are 4-dimensional: every leg carries both the ket and the bra index of ρ, i.e. the forward and backward branches of the Keldysh contour folded together. Then
$$\langle A_x \rangle_{N\delta t} = \langle \mathbb{1} |\, A_x\, (\mathcal{D}\mathcal{G})^N\, |\rho_0\rangle$$
is a two-dimensional network of N time layers by L sites, closed by the product state ρ_0 at the bottom and by the trace $\langle \mathbb{1}|$ (identity on every site except x) at the top.
- *Light cone.* A unitary channel acting on identities returns identities, and the dephasing does not change $\mathbb{1}$ (trace preservation, unitality). Gates outside the backward light cone of the site x and the final time therefore cancel, and only a cone of width $\approx 2vN\delta t$ contributes.
- *Transverse contraction.* Grouping the network into columns, one per site (or unit cell), each column is a spatial transfer matrix T_j acting on time-like vectors of dimension 4^N, the full history of the boundary between two sites. For a translation-invariant infinite chain, $\langle A_x \rangle = \langle \mathrm{IM}_L |\, T_x[A]\, |\mathrm{IM}_R\rangle$, where the influence matrices $\langle \mathrm{IM}_L|$, $|\mathrm{IM}_R\rangle$ are the dominant (fixed-point) eigenvectors of T from the left and from the right. They are found by applying T repeatedly to a boundary vector and compressing the result as an MPS *in time*: the transverse contraction. With site-dependent couplings (our random c) the fixed point is replaced by the product of the transfer matrices of the actual sites, applied from the boundaries.
- *Meaning and cost.* The influence matrix is the Feynman–Vernon influence functional of the rest of the chain on the chosen site: it encodes all memory effects of the many-body environment on the local dynamics. Its cost is set by its *temporal entanglement* (the entanglement of the IM as an MPS across a time cut), not by the spatial entanglement of $\rho(t)$. For a site coupled to a perfectly dephasing (Markovian) environment the IM is a product in time. Strong local dephasing lowers temporal entanglement, and the approach reaches $L = \infty$ directly. That is what the finite-size cap of Sec. 7 prevents with exact methods at small γ.

CLIK: eigenvalues from time-evolved states. CLIK (Westhoff, Moroder, Schollwöck and Paeckel) computes slow eigenvalues of $\mathcal{L}$ without diagonalizing it. By Eq. (7), the evolved states $\rho_n = e^{\mathcal{L}z_n}\rho_0$, sampled at $z_n = n\,\delta t$, contain each mode R_k with weight $e^{\lambda_k z_n}$: the slow modes survive, and the span of the samples approximates the slow invariant subspace (a Krylov-like space

generated by $e^{\mathcal{L}\delta t}$). On this span $\mathcal{L}$ is projected (Rayleigh–Ritz, Galerkin): writing $r = \sum_n c_n \rho_n$ and demanding $\langle \rho_m, (\mathcal{L} - \lambda) r \rangle = 0$ for all m gives the generalized eigenproblem

$$\mathsf{L}\, c = \lambda\, \mathsf{M}\, c, \qquad \mathsf{M}_{mn} = \langle \rho_m, \rho_n \rangle, \qquad \mathsf{L}_{mn} = \langle \rho_m, \mathcal{L}\rho_n \rangle, \tag{40}$$

solved after removing the near-zero eigenvalues of the Gram matrix M (the samples are nearly linearly dependent). Two evolutions are combined: a physical state (Tr = 1) in real time, for the steady state and slow modes, and a traceless state along the complex contour $z = t\, e^{-i\alpha}$. Along it mode k is damped at $\xi_k = -\operatorname{Re}\lambda_k \cos\alpha - \operatorname{Im}\lambda_k \sin\alpha$, so modes with $\operatorname{Im}\lambda_k > 0$ are enhanced, provided $\alpha < \alpha_{\max} = \min_k(\varphi_k - \pi/2)$ ($\lambda_k = |\lambda_k| e^{i\varphi_k}$), beyond which some mode grows. Left eigenvectors come from the same construction with $\mathcal{L}^\dagger$ (backward evolution, Sec. 2.1). With exact states the method is limited only by the time needed to separate the modes ($T \sim$ a few $/\Delta$, Δ the gap between the wanted modes and the rest); with MPS states the span has an effective bond dimension of order $N_K \chi$, but every state carries its truncation error, which sets the noise floor of the Gram matrix. Implementation: `julia/clik.jl` (dense backend for $L \leq 6$ with exact propagators, MPS backend with two-site TDVP and global expansion), following `data/docs/CLIK_MPS/IMPLEMENTATION.md`.

Decay rates are defined in Sec. 4.

4 How the decay rates are computed

All rates in Secs. 6–8 are obtained the same way, from the time series of one signal $S(t)$ per run: $m_\perp$ for the scar, $s_\perp$ for the x-Néel state, $|s_z|$ for the z-Néel state (Table 1). All three start at $S(0) = 1$. Code: `rates` in `plot_scripts/fit_states.py` and `rate` in `fit_small_gamma.py`.

Envelope. The signal precesses (the scar's $m_\perp$ is smooth, but s_z crosses zero, and small systems show revivals), so the rate is read from the non-increasing envelope

$$E(t) = \max_{t' \geq t} S(t'), \tag{41}$$

computed on the output grid ($dt = 0.1$, or 0.25 for the small-γ jump runs).

The $1/e$ rate Γ_e (main rate). t_e is the first time with $E(t_e) < 1/e$. Between the two output times t_{i-1}, t_i that bracket the crossing it is interpolated linearly in $\ln E$:

$$t_e = t_{i-1} + (t_i - t_{i-1}) \frac{\ln E_{i-1} + 1}{\ln E_{i-1} - \ln E_i}, \qquad \Gamma_e = \frac{1}{t_e}. \tag{42}$$

This is *not* a fit of $A_0 e^{-\Gamma t}$: there is no free amplitude and no fit window. It is the rate of the exponential that starts at the true initial value $S(0) = 1$ and passes through the measured envelope at $1/e$. For a single exponential $e^{-\Gamma t}$ it returns Γ exactly. For a general decay, write $E(t) = \exp[-\int_0^t \lambda(t')\, dt']$ with the instantaneous rate $\lambda = -d \ln E/dt$ (this uses $E(0) = 1$). The crossing condition $\ln E(t_e) = -1$ reads $\int_0^{t_e} \lambda\, dt = 1$, so

$$\Gamma_e = \frac{1}{t_e} = \frac{1}{t_e} \int_0^{t_e} \lambda(t)\, dt, \tag{43}$$

the instantaneous rate averaged from $t = 0$ to the $1/e$ time. This matters here because the decays are not single exponentials. The scar decays multi-exponentially, and at small γ it follows $\exp[-\gamma \int \bar{D}]$, whose instantaneous rate grows from 2γ to γD_{sat} (Sec. 8). A fit of $A_0 e^{-\Gamma t}$ with A_0 free would instead depend on the window (Appendix A): with an early fast drop it gives $A_0 < 1$ and the slower late rate. The single-kick prediction of Γ_e uses the same definition, $\gamma \int_0^{t_e} \bar{D} = 1$.

Statistical error (trajectories). With $\sigma_m(t)$ the standard error of the trajectory-averaged signal (from the errors of m_x, m_y or s_x, s_y), a shift δE of the envelope at the crossing moves the crossing time by $\delta t = \delta E/|dE/dt|$, so $\sigma_t = \sigma_m(t_e)/|dE/dt|$, with the slope taken over ± 4 output steps. From $\Gamma_e = 1/t_e$, $\delta\Gamma_e = \delta t/t_e^2$, i.e. $\sigma_\Gamma = \Gamma_e^2\sigma_t$. For an exponential near t_e, $|dE/dt| = \Gamma_e/e$ there, so $\sigma_\Gamma/\Gamma_e = e\,\sigma_m$. The exact evolution has no statistical error.

Tail rate Γ_{tail}. A least-squares straight-line fit of $\ln E$ between the times where E drops below 10^{-2} and 10^{-3}. This one *is* an $A_0 e^{-\Gamma t}$ fit, with A_0 free, on the tail only; it measures the slowest modes (the tower edges for the scar). It is used for exact runs only: trajectory runs reach their statistical floor in that window. Runs stop once the signal stays below $\max(10^{-3}, 3\sigma)$ for 10 time units, or at a fixed $t_{\max}$ (the small-γ runs, $t_{\max} = 3/(\gamma(L-1))$, past t_e).

Thermodynamic rate Γ_∞. For every state and γ, $\Gamma_e(L) = A + B/L$ is fitted (unweighted) on $L \geq 8$ and $\Gamma_\infty = A$. The error is the larger of the fit error and $|A - A_{\text{quad}}|$, with A_{quad} the intercept of a quadratic fit in $1/L$ on $L \geq 6$. Three groups of sizes and γ are handled differently:

- **Néel states with finite-size revivals.** The affected small sizes are left out: x-Néel $L = 6$ at $\gamma \leq 0.05$, z-Néel $L = 6, 8$ at $\gamma \leq 0.02$. z-Néel has no quadratic fit; its error is $|A - \Gamma_e(L_{\max})|$.
- **Scar at $\gamma < 0.01$.** Every size sits at the cap $\Gamma_e \approx \gamma(L-1)$ (Sec. 7), so the $L \geq 8$ fit is not used. Instead:
  - Γ_e of the largest size is a lower bound on Γ_∞, since Γ_e grows with L there.
  - A second fit $A + B/L$ on $L = 14$–20 only gives an estimate. Its error is the larger of the fit error and the change of A when $L = 12$ is added (Table 6).
  - The collapse $\Gamma_e/\gamma = \xi f(L/\xi)$, $\xi = \gamma^{-\nu}$, gives the power law $\Gamma_\infty \propto \gamma^{1-\nu}$.
- **$\gamma = 0$.** It is not fitted. Its Γ_e is the mean over the fitted sizes; the scar never reaches $1/e$, so its Γ_e is 0.

Figure 1 shows the definition at work on the largest system, $L = 20$: at $\gamma = 0$ the precession is perfect and $\Gamma_e = 0$; for $\gamma > 0$ the exponential $e^{-\Gamma_e t}$ starts at 1, meets the envelope at $1/e$, and bounds the oscillation of m_x, while the free-spin $e^{-2\gamma t}$ is far slower at small γ. The envelope is not a single exponential: at small γ it lies above $e^{-\Gamma_e t}$ before t_e (by up to ≈ 0.1 at $\gamma = 0.005$) and close to it or below after, because its instantaneous rate grows from 2γ to γD_{sat} (Sec. 8); Γ_e is the mean of that rate up to t_e. At $\gamma = 1$ it is almost a single exponential, close to $e^{-2\gamma t}$.

5 Exact results

These hold for any coefficients.

Dark states and steady states. $|\Phi_0\rangle = |\uparrow \cdots \uparrow\rangle$ and $|\Phi_L\rangle = |\downarrow \cdots \downarrow\rangle$ are eigenstates of H and of every σ_j^z, hence dark states of the dissipator. The kernel of $\mathcal{L}$ is spanned by

$$\mathbb{1}, \qquad |\Phi_0\rangle\langle\Phi_0|, \qquad |\Phi_L\rangle\langle\Phi_L|, \tag{44}$$

and ED finds exactly three zero eigenvalues for $L = 4, 5, 6$. The infinite-temperature state is therefore *not* the unique steady state. The coherence $|\Phi_0\rangle\langle\Phi_L|$ is an eigenmode with $\lambda = -2\gamma L - ihL$, which saturates the general bound $-2\gamma L \leq \operatorname{Re}\lambda \leq 0$.

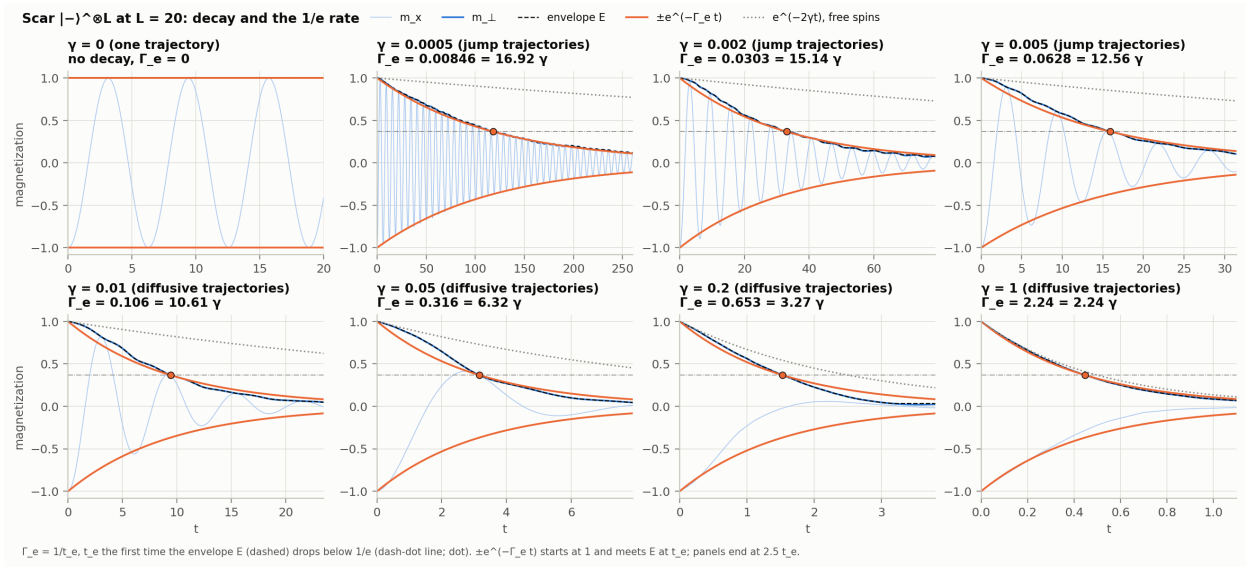


Figure 1: Scar at $L = 20$: m_x (light), $m_\perp$ (blue) and its envelope E (dashed) for $\gamma = 0$ and seven $\gamma > 0$ (jump trajectories for $\gamma < 0.01$, diffusive above; $\gamma = 0$ is one deterministic trajectory, run to $t = 20$). Orange: $\pm e^{-\Gamma_e t}$; dot: $(t_e, 1/e)$; dotted: $e^{-2\gamma t}$. Each panel ends at $2.5\,t_e$ (`plot_scar_decay_L20.py`).

Edge coherences of the tower. Since $\sigma_j^z|\Phi_0\rangle = |\Phi_0\rangle$ and $\sum_j \sigma_j^z = 2S^z_{\rm tot}$,

$$\mathcal{D}\big(|\Phi_0\rangle\langle\Phi_n|\big) = \gamma\,|\Phi_0\rangle\langle\Phi_n|\Big(\sum_j \sigma_j^z - L\Big) = -2n\gamma\,|\Phi_0\rangle\langle\Phi_n|, \tag{45}$$

and both states are eigenstates of H, so $|\Phi_0\rangle\langle\Phi_n|$ (and likewise $|\Phi_L\rangle\langle\Phi_{L-n}|$) is an exact Liouvillian eigenmode with

$$\lambda = -2n\gamma - ihn \qquad \text{for every } L. \tag{46}$$

The slowest tower modes therefore decay at exactly 2γ, independent of L. In first-order perturbation theory the coherences $|\Phi_n\rangle\langle\Phi_{n+1}|$ decay at $\gamma L(1 - m_n m_{n+1})$ with $m_n = 1 - 2n/L$: 2γ at the edges, $\approx \gamma L$ in the middle of the tower. The scar signal is a binomially weighted sum of these, which makes it multi-exponential.

Initial loss of tower weight. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = |-\rangle^{\otimes L}$, $w = \mathrm{Tr}(P_{\rm tower}\rho)$ and $[P_{\rm tower}, H] = 0$,

$$\dot w(0) = \gamma \sum_j \big(\|P_{\rm tower}\sigma_j^z|\psi\rangle\|^2 - 1\big), \qquad P_{\rm tower}\sigma_j^z|\psi\rangle = \tfrac{2}{L}S^z_{\rm tot}|\psi\rangle \;\Rightarrow\; \dot w(0) = -\gamma(L-1), \tag{47}$$

using permutation symmetry of the tower and $\langle (S^z_{\rm tot})^2\rangle = L/4$. A kick keeps a fraction $1/L$ of the state in the tower; the rest is $k \neq 0$ spin waves. Reproduced exactly by the numerics for $L = 4, \ldots, 8$. At late times $w \to (L+1)/2^L$.

Overlaps. $P_{\rm tower}\sigma_j^\alpha P_{\rm tower} = \frac{2}{L}S^\alpha_{\rm tot}P_{\rm tower}$ for every j, so the staggered operators $O^\alpha_\pi = \sum_j (-1)^j\sigma_j^\alpha$ have zero tower part. The x-Néel signal (Sec. 6) is therefore blind to the tower.

6 Scar against thermal decay up to $L = 20$

6.1 States and signals

The z-Néel state is a biased reference: it is diagonal in the z basis, so $\mathcal{D}(\rho_0) = 0$, and s_z is invisible to the bath, $\mathcal{D}^\dagger(\sigma_j^z) = 0$. The bath acts only after H_I has rotated weight into σ^x, σ^y: with $H_I = 0$,

state	ρ_0	signal	at $H_I = 0$
scar	$\|-\rangle^{\otimes L}$	$m_\perp = \sqrt{m_x^2 + m_y^2}$	$e^{-2\gamma t}$
x-Néel (thermal reference)	$\|+-+-\cdots\rangle$, even L	$s_\perp = \sqrt{s_x^2 + s_y^2}$	$e^{-2\gamma t}$
z-Néel	$\|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$, even L	$\|s_z\|$	no decay

Table 1: Initial states, $m_\alpha = \frac{1}{L}\sum_j \langle\sigma_j^\alpha\rangle$, $s_\alpha = \frac{1}{L}\sum_j (-1)^j \langle\sigma_j^\alpha\rangle$.

s_z never decays, and at large γ it is Zeno-protected. The x-Néel $s_\perp$ has the same free-spin baseline $e^{-2\gamma t}$ as the scar's $m_\perp$ and zero tower overlap, so it is the like-for-like thermal reference. Neither Néel state is a dark state of $\mathcal{L}$.

Runs. Rates as in Sec. 4. The scar is computed for $L = 4, \ldots, 12$ and 14 exactly and $L = 16, 18, 20$ with trajectories; the Néel states for even L (exact up to 14 for x-Néel and up to 12 for z-Néel, trajectories $L = 16, 18, 20$ for x-Néel). The γ grid is $0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10$. Where exact and trajectory runs overlap ($L = 12, 14$), Γ_e agrees to 1–2.5% for $\gamma \leq 2$.

6.2 Scar

At $\gamma = 0$ the scar state precesses perfectly, $\langle\sigma_j^x\rangle = -\cos ht$, $\langle\sigma_j^y\rangle = -\sin ht$, on every site and for every L ($\lesssim 10^{-13}$). With dephasing, $m_\perp$ decays faster than the free-spin $e^{-2\gamma t}$ (Fig. 2). Table 2 gives Γ_e/γ.

γ	$L = 8$	10	12	14	16	18	20
0.01	6.73	7.84	8.49	8.88	9.56	10.24	10.61
0.02	6.00	6.77	7.77	8.15	8.34	8.50	8.64
0.05	5.76	6.03	6.24	6.25	6.26	6.28	6.32
0.1	4.36	4.45	4.48	4.47	4.43	4.46	4.51

Table 2: Scar Γ_e/γ (exact for $L \leq 14$, trajectories for $L \geq 16$). $\gamma = 0.2$: 3.0–3.3; $\gamma = 1$: 2.21–2.26; $\gamma = 10$: ≈ 2.0.

For $\gamma \geq 0.05$, Γ_e is L-independent from $L \approx 10$ to 20 (within $\approx 1\%$); at $\gamma = 0.02$ it flattens from $L \approx 16$; at $\gamma = 0.01$ it still grows at $L = 20$. At fixed L and $\gamma \to 0$, $\Gamma_e/\gamma \to L - 1$ (Sec. 7), so a size L shows the thermodynamic rate only if $L - 1 \gg \Gamma_\infty/\gamma$. The extrapolated rates are in Table 3 and Figs. 6, 7. The local exponent $d\ln\Gamma_\infty/d\ln\gamma$ is 0.73 (0.01–0.02), 0.50 (0.02–0.05), 0.44 (0.05–0.1), 0.51 (0.1–0.2), then 0.7–0.9 $\to 1$ at large γ: the lifetime grows as $\gamma^{-1/2}$ over $0.02 \leq \gamma \leq 0.2$. The steeper step at 0.01–0.02 rests on the least converged point: at $\gamma = 0.01$ the fit on $L \geq 14$ gives 0.147 ± 0.009 instead of 0.128 (Sec. 7, Table 6), which brings the exponent there to 0.41. At large γ, $\Gamma_\infty \to 2\gamma$: the dephasing outruns H_I and every spin dephases on its own.

γ	0.01	0.02	0.05	0.1	0.2	0.5	1	2	5	10
scar Γ_∞	0.128(12)	0.213(10)	0.335(14)	0.453(10)	0.647(22)	1.24(2)	2.26(1)	4.22(8)	9.5(5)	19.7(3)
scar Γ_∞/γ	12.8	10.6	6.7	4.5	3.2	2.5	2.26	2.1	1.9	1.97
x-Néel Γ_∞	4.74(36)	4.75(38)	4.78(34)	4.90(13)	4.99(11)	5.32(11)	5.78(8)	7.04(15)	11.3(3)	20.6(3)
z-Néel Γ_∞	5.04(9)	5.03(9)	5.08(19)	4.89(7)	4.72(5)	4.28(2)	3.77(3)	2.91(2)	1.71	0.98

Table 3: Thermodynamic rates. At $\gamma = 0$: scar 0 (perfect revivals), x-Néel 4.76 ± 0.08 (mean Γ_e over $L \geq 8$), z-Néel 4.96 ($L = 10, 12$). z-Néel from $L \leq 12$ only.

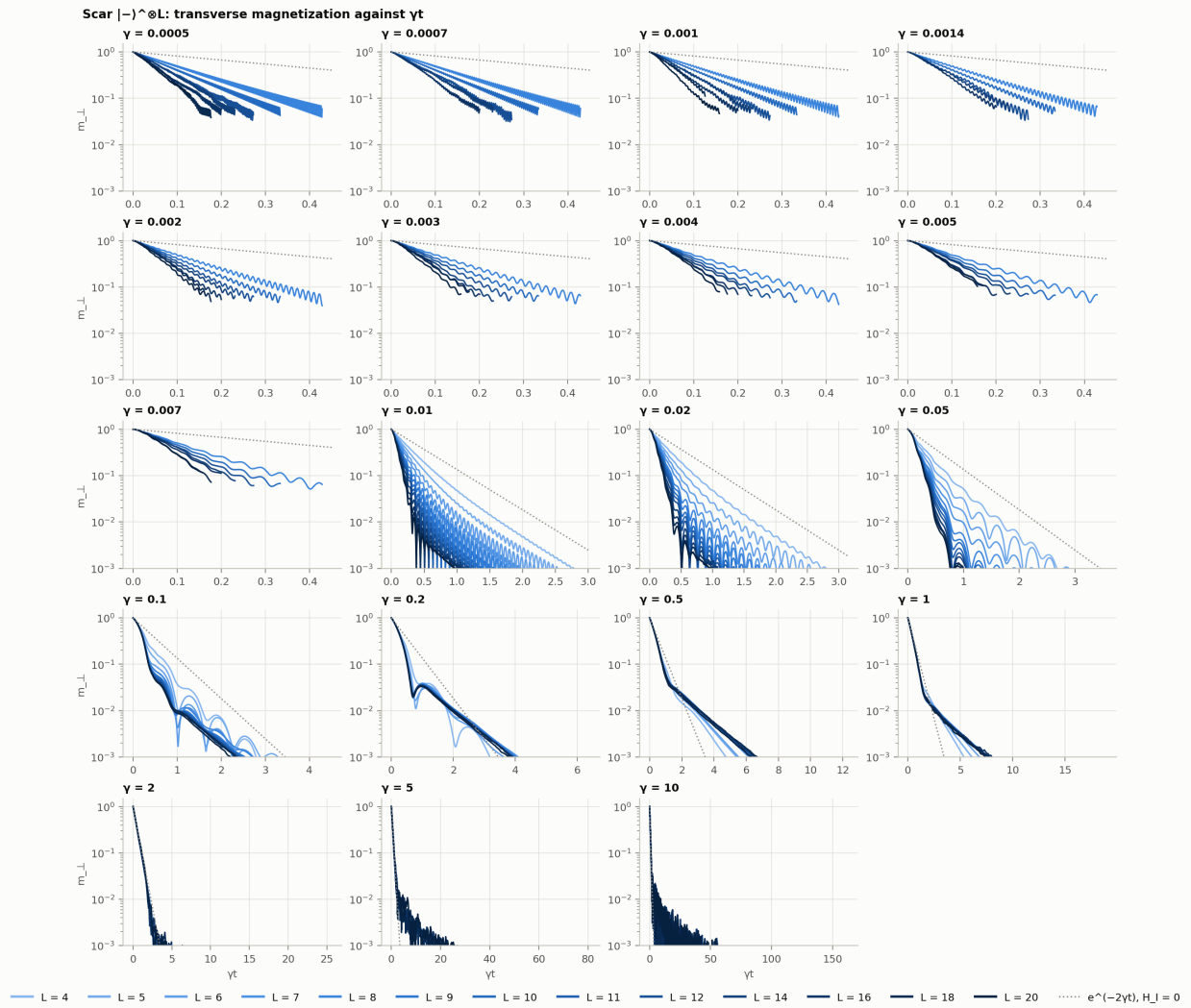


Figure 2: Scar: $m_\perp$ against γt for every γ and L; dotted: $e^{-2\gamma t}$. The panels with $\gamma < 0.01$ are the small-γ runs of Sec. 7, which stop at $t = 3/(\gamma(L-1))$ (`plot_scripts/plot_states.py`).

6.3 Thermal references

x-Néel. At $\gamma = 0$ it thermalizes with $\Gamma_e \approx 4.8$ for every $L \geq 8$ ($L = 4, 6$ have finite-size revivals and never reach $1/e$ at $\gamma \leq 0.05$). Γ_e is L-independent for $L \geq 8$ at every γ, within $\approx 1\%$ (the $L = 16$ realization lies $\approx 4\%$ lower at every γ, also at $\gamma = 0$: a property of its c). Γ_∞ stays at the closed-system rate for $\gamma \leq 0.1$ and then approaches 2γ ($+0.6$ at $\gamma = 10$): there is no Zeno slowdown (Fig. 3).

z-Néel. At $\gamma = 0$, $|s_z|$ decays at $\Gamma_e \approx 5$ for $L \geq 10$ ($L = 4$ has strong revivals). $\Gamma_\infty \approx 5.0$ for $\gamma \leq 0.05$, then decreases, $\propto 1/\gamma$ at large γ (0.98 at $\gamma = 10$): a Zeno slowdown, as expected for a signal the bath cannot see directly (Fig. 4). The number of down spins (Fig. 5) rises at $\gamma = 0$ from $L/2$ to $\approx 0.55L$ ($m_z \approx -0.1$, the thermal value fixed by the energy since H contains $hS^z_{\rm tot}$), and with dephasing overshoots and relaxes back to $L/2$ at ≈ 1.5–2.3γ, L-independent for $L \geq 8$.

Scar against thermal. $\Gamma_\infty(\text{scar}) < \Gamma_\infty(x\text{-Néel})$ at every γ, by a factor 37 ($\gamma = 0.01$), 22 (0.02), 14 (0.05), 11 (0.1), 7.7 (0.2), 4.3 (0.5), 2.6 (1), 1.7 (2), 1.05 (10). The scar rate goes to 0 as $\gamma \to 0$; the thermal rate stays at ≈ 4.8 (Fig. 7). This is the expected contrast of the Introduction of the project: scar revivals are regular in the weak-dissipation limit, the thermal signal is not. The z-Néel rate crosses the scar rate at $\gamma \approx 1.5$, an artefact of its Zeno protection.

7 Small γ: the finite-size cap

To reach the regime $\gamma \to 0$, the scar was run at $\gamma = 0.0005$, 0.0007, 0.001, 0.0014, 0.002, 0.003, 0.004, 0.005, 0.007 with jump trajectories for $L = 12, \ldots, 20$ (without 0.0007, 0.0014 at $L = 18, 20$) and exactly for $L = 8, 10$ (every γ) and 12 ($\gamma \geq 0.002$), up to $t = 3/(\gamma(L-1))$. The traj-seeds of one (L, γ) are pooled, weighted by their trajectory counts.

Cross-checks. The jump trajectories have a large per-trajectory spread (≈ 0.3 at $m_\perp \approx 0.5$, against ≈ 0.08 for diffusive trajectories at $L = 12$): a trajectory is either undamaged or carries a damaged region. With 448 trajectories the statistical error of Γ_e is 1.5–6%, up to 7.5% at $L = 20$ with 224. All γ of one L use the same random numbers, so the kick sequences coincide up to the time scale $1/\gamma$ and the errors are correlated across γ. Table 4 compares jump runs with the exact evolution and with diffusive trajectories: the differences are within $\approx 2\sigma$ (Fig. 8); exact and diffusive agree with each other to 0.3%.

L	γ	reference	rms z	$\Gamma_e^{\rm jump}/\Gamma_e^{\rm ref} - 1$
12	0.002	exact	0.74	$+0.8\%$
12	0.003	exact	1.15	-7.1%
12	0.004	exact	1.04	$+2.2\%$
12	0.005	exact	1.83	-10.6%
12	0.007	exact	1.22	-5.1%
12	0.002	diffusive	0.71	$+0.4\%$
12	0.005	diffusive	1.72	-10.4%
14	0.002	diffusive	0.91	$+0.3\%$
14	0.005	diffusive	0.87	-6.4%

Table 4: Jump trajectories against the references: rms over t of $(m_\perp^{\rm jump} - m_\perp^{\rm ref})/\sigma$ while $m_\perp^{\rm ref} > 0.05$, and the difference of Γ_e (statistical error of the jump Γ_e: 4–6%).

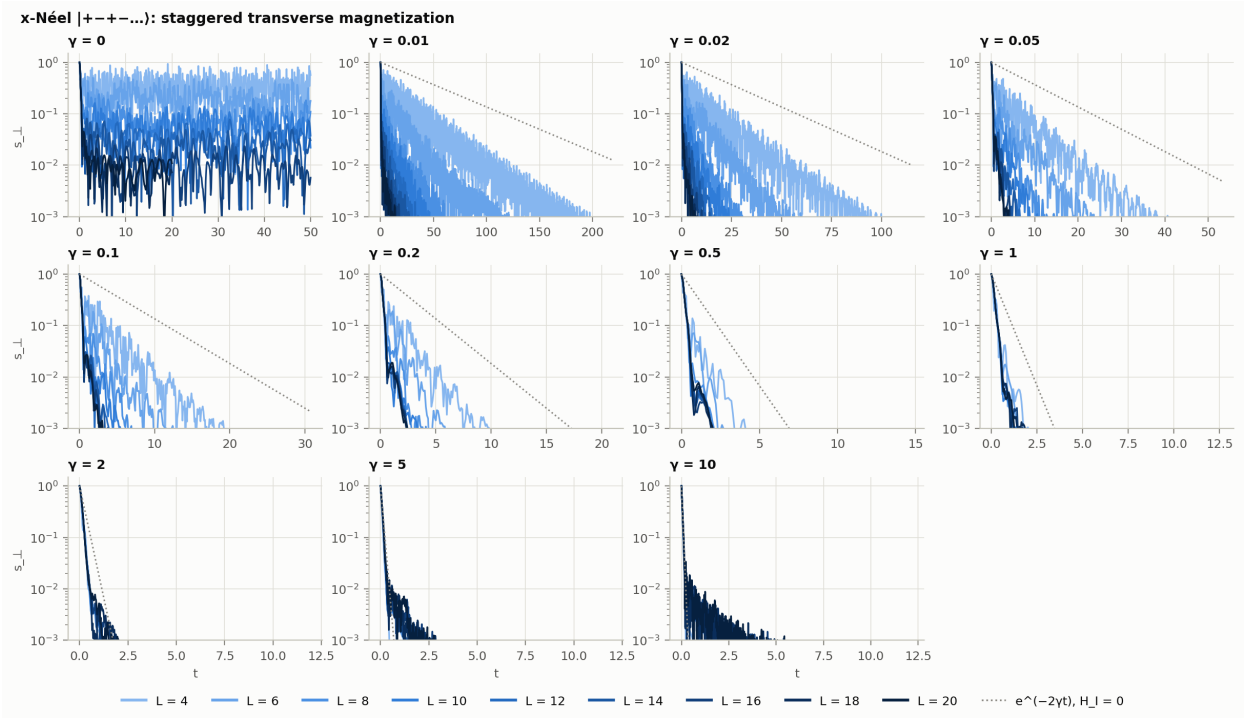


Figure 3: x-Néel: $s_\perp$ against t; dotted: $e^{-2\gamma t}$ (`plot_states.py`).

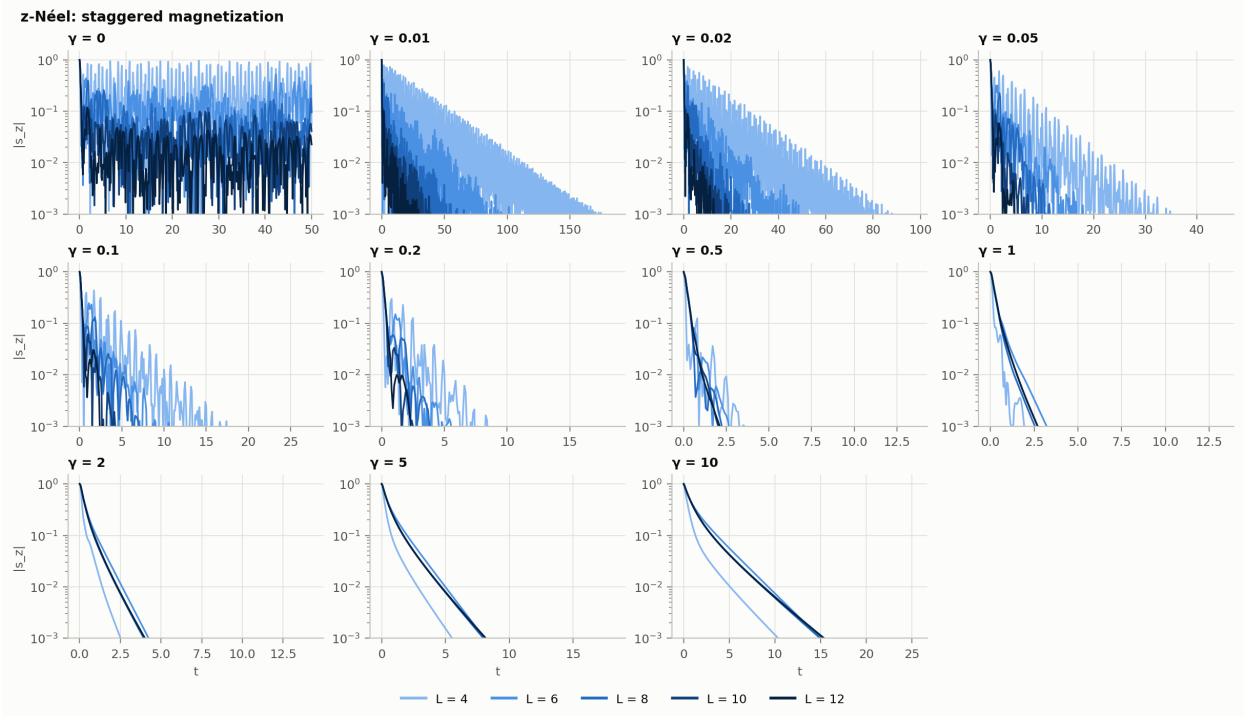


Figure 4: z-Néel: $|s_z|$ against t (`plot_states.py`).

Rates. Table 5 and Fig. 9; in Fig. 7 the largest size gives lower bounds on Γ_∞ (0.0085 at $\gamma = 0.0005$, 0.030 at 0.002, 0.084 at 0.007), all below the collapse power law and above 2γ. At $L \le 10$, $\Gamma_e/\gamma \approx L - 1$ for every $\gamma \le 0.003$ (6.8–7.0 at $L = 8$, 8.9 at $L = 10$); at $L = 12$–20 the smallest γ reach $L - 1$ within the errors (11.5 at $L = 12$, 14.6 at $L = 16$, 16.9–19.0 at $L = 20$). At fixed L, Γ_e/γ decreases with γ towards Γ_∞/γ. The cap equals the initial loss rate of the tower weight, Eq. (47): at $\gamma \to 0$ with L fixed, the local envelope decays like the global tower weight, because the damage of a kick has time to spread over the whole chain before the next kick (Sec. 8).

γ	$L = 8$	10	12	14	16	18	20
0.0005	6.82	8.89	11.46	11.82	14.60	14.65	16.92
0.0007	6.93	8.82	10.95	11.45	13.49	–	–
0.001	6.98	8.92	11.55	11.48	14.10	14.64	19.04
0.0014	6.81	8.86	10.71	10.69	12.71	–	–
0.002	6.81	8.93	10.24	11.67	13.25	13.34	15.14
0.003	6.85	8.91	10.06	11.10	11.94	13.09	15.30
0.004	6.88	8.37	9.89	10.56	11.73	11.84	13.21
0.005	6.80	8.31	9.31	9.92	11.39	11.66	12.56
0.007	7.17	7.99	8.75	9.23	10.08	11.92	12.01
0.01	6.73	7.84	8.49	8.88	9.56	10.24	10.61
$L - 1$	7	9	11	13	15	17	19

Table 5: Scar Γ_e/γ at small γ: exact for $L \le 12$ ($L = 12$ only $\gamma \ge 0.002$), jump trajectories otherwise; $\gamma = 0.01$ from Table 2.

Collapse. If the finite-size effects are set by one length $\xi(\gamma) = \gamma^{-\nu}$, then $\Gamma_e/\gamma = \xi f(L/\xi)$ with $f(x) \approx x$ at small x (the cap) and $f \to$ const at large x, so that $\Gamma_\infty \propto \gamma^{1-\nu}$. Minimizing the spread of $(\Gamma_e/\gamma)\gamma^\nu$ at common $L\gamma^\nu$ ($\gamma \le 0.05$, $L \ge 8$) gives $\nu = 0.33$, with a cost within a factor 2 of the minimum for $\nu = 0.24$–0.48 (the cost at $\nu = 0.5$ is 2.2× the minimum). Ballistic spreading of the damage ($\xi \sim vt$, $\nu = 1/2$, lifetime $\propto \gamma^{-1/2}$) and diffusive spreading ($\nu = 1/3$, lifetime $\propto \gamma^{-2/3}$) are both within this range: $L \le 20$ does not separate them.

Extrapolation from the large sizes. The fit $A + B/L$ on $L \ge 8$ (Sec. 6) includes sizes that are still at the cap when γ is small. A second fit uses only $L = 14, 16, 18, 20$. Its error is the larger of the fit error and the change of A when $L = 12$ is added. It needs at least three sizes, so $\gamma = 0.0007$ and 0.0014 (only up to $L = 16$) are left out. Results are in Table 6 and Fig. 10. They appear as the $L = \infty$ points of Fig. 9 and as open squares in Fig. 7.

- At $\gamma \ge 0.05$ the two fits agree.
- At $\gamma = 0.01$ and 0.02 they differ by 15% and 8%, with opposite signs: the $L \ge 8$ value at $\gamma = 0.01$ (0.128) is pulled down by the capped small sizes.
- At $\gamma < 0.01$ the extrapolated $A/\gamma = 18$–34 lies 40–80% above $\Gamma_e(L = 20)/\gamma$. A moves by up to 25% when $L = 12$ is added, so these values are estimates, not converged limits. They sit inside the band of the collapse power law.

A power law $A \propto \gamma^p$ fitted to all ten values ($0.0005 \le \gamma \le 0.05$) gives $p = 0.58 \pm 0.01$ (weighted) or 0.66 ± 0.04 (unweighted), i.e. $\nu = 1 - p = 0.34$–0.42, consistent with the collapse. Restricted to $\gamma \le 0.007$ it gives $p = 0.80 \pm 0.06$. That steepening is expected from the fit itself: at smaller γ the sizes are closer to the cap, so the extrapolation underestimates Γ_∞ more.

γ	0.0005	0.001	0.002	0.003	0.004	0.005	0.007	0.01	0.02	0.05
A/γ, $L \geq 14$	27.4	33.8	22.1	24.0	18.5	18.3	19.4	14.8	9.8	6.5
error (%)	11	19	9	11	8	6	10	6	2	1
A/γ, $L \geq 12$	24.3	27.4	21.6	21.6	17.5	17.3	17.4	13.8	9.9	6.4
$\Gamma_e(L=20)/\gamma$	16.9	19.0	15.1	15.3	13.2	12.6	12.0	10.6	8.6	6.3
Γ_∞/γ, $L \geq 8$ (Table 3)	–	–	–	–	–	–	–	12.8	10.6	6.7

Table 6: Scar: $A + B/L$ on the large sizes ($L = 14, \ldots, 20$), with the check on $L \geq 12$ (`fit_small_gamma.py`, `states_inverse_L_large.csv`).

8 Single-kick test

Model. Take a kick σ_k^z at time t' on the otherwise noiseless scar state, and evolve it with H for a time τ. With $c_j(\tau) = (\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle)$ divided by its noiseless value at $t' + \tau$, define the damage

$$D(\tau) = \sum_j \big[1 - \mathrm{Re}\, c_j(\tau)\big], \qquad D(0) = 2, \tag{48}$$

and its average $\bar{D}(\tau)$ over 16 phases $\varphi = ht'$ and the kicked sites.

Derivation of the prediction. Write the transverse magnetization as the complex number $m_+ = m_x + im_y = L^{-1}\sum_j(\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle)$. Its noiseless value is $-e^{iht}$ (reference 1), so after one kick

$$\frac{m_+(t)}{-e^{iht}} = \frac{1}{L}\sum_j c_j(t-t') = 1 - \frac{D(t-t')}{L} + i\,\frac{1}{L}\sum_j \mathrm{Im}\, c_j(t-t'),$$

and, to first order in $1/L$, $m_\perp = |m_+| = 1 - D(t-t')/L$: one kick at t' multiplies $m_\perp$ by $f(t-t') = 1 - D(t-t')/L$. Two assumptions turn this into a prediction for the Lindblad dynamics:

1. *independent kicks:* with kicks at $t_1, \ldots, t_N$, the factors multiply, $m_\perp(t) \approx \prod_{i=1}^N f(t-t_i)$, i.e. the damaged regions of different kicks do not overlap;

2. *phase average:* each factor is replaced by its average over the kicked site and over the phase $\varphi = ht_i$ of the kick, $f \to 1 - \bar{D}/L$ (the 16 phases sample one precession period).

In the jump unravelling (Sec. 3) the kick times form a Poisson process of rate γL on $[0, t]$, independent of the state. For a Poisson process of rate r and any function f, $\mathbb{E}\big[\prod_i f(t_i)\big] = \exp\big[-r\int_0^t \big(1 - f(t')\big)\,dt'\big]$ (sum over the number of kicks, each uniform in $[0,t]$: $\sum_N e^{-rt}\frac{(rt)^N}{N!}\big(\frac{1}{t}\int_0^t f\big)^N$). With $r = \gamma L$ and $1 - f = \bar{D}/L$,

$$m_\perp(t) \approx \exp\Big[-\gamma L\int_0^t \frac{\bar{D}(t-t')}{L}\,dt'\Big] = \exp\Big[-\gamma\int_0^t \bar{D}(\tau)\,d\tau\Big], \tag{49}$$

independent of L except through $\bar{D}$. For free spins ($H_I = 0$) a kick reverses one spin and nothing else, $D \equiv 2$ exactly, the kicks are exactly independent, and Eq. (49) gives the exact $e^{-2\gamma t}$. The instantaneous decay rate of the prediction is $\gamma\bar{D}(t)$: 2γ at $t = 0$, γD_{sat} once the damage has saturated. The predicted Γ_e solves $\gamma\int_0^{t_e}\bar{D} = 1$, with no free parameter. Runs: all sites for $L = 12, \ldots, 20$ up to $\tau = 25$, and four sites $(0, L/4, L/2, 3L/4)$ for $L = 22, 24$ up to $\tau = 8$. Beyond $\tau = 25$, $\bar{D}$ is continued with its mean over $20 \leq \tau \leq 25$.

Damage. Fig. 11. $\bar{D}$ jumps from 2 to ≈ 5.9 within $\tau = 1$ for every L, then grows, and saturates with slow oscillations (period ≈ 6). At $\tau = 4$: $9.10, 9.99, 10.66, 11.02, 11.35, 11.49, 11.60$ for $L = 12, 14, \ldots, 24$; at $\tau = 8$: $9.73, 11.02, 12.36, 13.44, 14.62, 15.54, 16.32$. The saturation value is $11.0, 12.4, 14.0, 15.4, 17.1$ for $L = 12, \ldots, 20$, i.e. $\approx L - 1$ at $L = 12$ and still 0.6–1.9 below $L - 1$ at $\tau = 25$ for the larger sizes. The size dependence sets in at $\tau \approx 3$ ($L = 12$) to $\tau \approx 6$ ($L = 22$ against 24). For $L = 22, 24$, $\bar{D}$ grows by ≈ 1.0–1.2 per unit of τ between $\tau = 4$ and 8, roughly linearly (only four kicked sites are sampled; the spread over the 64 runs is ≈ 1.2 at $\tau = 8$).

Finite L. The prediction (49) follows the whole measured time series (Fig. 12). Over all 55 points ($L = 12, \ldots, 20$, $\gamma \leq 0.05$), the ratio of measured to predicted Γ_e is 0.90–1.19, mostly 0.95–1.05 (Fig. 13, left). At $\gamma < 0.01$ this is within the statistical errors. At $\gamma = 0.01$–0.05, where the errors are below 1%, the ratio is systematically 0.95–1.00: the independent-kick formula counts overlapping damage twice, so it predicts a slightly faster decay. So at small γ the scar decay is fully set by isolated kicks, and the finite-size cap $\Gamma_e/\gamma \to L - 1$ is the saturated damage $D_{\text{sat}} \approx L - 1$.

Thermodynamic limit. Fig. 13, right: Γ_∞ against the prediction from $\bar{D}$ at $L = 24$ ($\tau \leq 8$, so $\gamma \geq 0.02$). The two agree at $\gamma = 0.05$ (0.332 against 0.335 ± 0.014). Above that, kicks overlap within the damage time, and the prediction is too fast: 0.53 against 0.45 at $\gamma = 0.1$, 0.84 against 0.65 at 0.2, 1.54 against 1.24 at 0.5. At $\gamma \geq 2$ both go to 2γ. At $\gamma = 0.02$ the prediction (0.181) is 15% below the $L \geq 8$ extrapolation (0.213 ± 0.010), but only 7% below the fit on $L \geq 14$ (0.196 ± 0.003, Table 6), and close to the largest measured size ($\Gamma_e(L = 20) = 0.173$). The remaining gap is either $\bar{D}$ still growing beyond $L = 24$ at $\tau \approx 5$, or the extrapolation.

Consequence. In the thermodynamic limit at small γ, Eq. (49) makes the lifetime a property of $\bar{D}_\infty(\tau)$ alone. Linear growth $\bar{D} \simeq 2v\tau$ (ballistic spreading of the damage) gives $\gamma v t_e^2 = 1$, i.e. $\Gamma_\infty = (\gamma v)^{1/2}$. Diffusive growth $\bar{D} \propto \tau^{1/2}$ gives $\Gamma_\infty \propto \gamma^{2/3}$. If the damage stays bounded, $\Gamma_\infty \propto \gamma$. The measured Γ_∞ ($\Gamma_\infty^2/\gamma \approx 2.0$–$2.3$ for $0.02 \leq \gamma \leq 0.1$) and the roughly linear growth of $\bar{D}$ at $L = 22, 24$ for $4 \leq \tau \leq 8$ favour the ballistic case, but the collapse admits both. Deciding needs $\bar{D}$ at longer τ in larger systems.

9 Next calculations: the lean data plan

The plan (`lean-data-plan.md`, 25 October 2026) aims at better data at minimum cost. It addresses five open points: (i) Γ_∞ at $\gamma = 0.01$–0.02 is the least converged (at $\gamma = 0.01$, Γ_e still grows at $L = 20$, and the fits on $L \geq 8$ and $L \geq 14$ differ by 15%); (ii) the growth of $\bar{D}(\tau)$ at $L \geq 24$ and $\tau > 8$, which fixes Γ_∞ at small γ through Eq. (49); (iii) there is one disorder realization per L; (iv) new observables: entanglement, operator entanglement of ρ, operator size and support (Sec. 3.9); (v) other single-site noise channels (Sec. 3.8). Existing data are never rerun. Costs are in Leonardo node-hours (NH; 1 NH = 112 core-hours).

Rules for every new run: stop at $t \approx 1.5\, t_e$, which is enough for Γ_e (the tail is only needed from exact runs); diffusive rather than jump trajectories at small γ; the smallest number of trajectories and of single-flip phases and sites that gives $\leq 2\%$ on the rates (set by step A below); the exact runs at $L \leq 10$ and the trajectories at $L = 16$ on the workstation *obelix*.

9.1 Step A: what the existing data say (no new runs)

Computed by `plot_scripts/lean_plan_analysis.py` from the data of Secs. 7–8.

A1. Lindblad data against the single-kick damage. By Eq. (49), $F(t) \equiv -\gamma^{-1} \ln m_\perp(t)$ should equal $\int_0^t \bar{D}(\tau)\, d\tau$ for every γ. Figure 14 overlays F for all $\gamma \leq 0.02$ at $L = 16, 18, 20$ with $\int \bar{D}$ of the same L: the curves fall on it, wiggles included. For the diffusive runs ($\gamma = 0.01$, 0.02), $F/\int \bar{D} = 0.90$–0.98 at $t = 4$–12, slightly below 1 (the overlap of kicks); the jump runs scatter by $\pm 25\%$, their noise amplified by $1/\gamma$. So the single-kick formula holds for the whole time dependence, not only for Γ_e.

The growth slope $s = d\bar{D}/d\tau$ can *not* be read from the Lindblad data at $L \leq 20$. The damage of a kick reaches around the ring at $t \approx L/s$ ($s = 1.16$ from $\bar{D}$ of $L = 24$ on $\tau = 4$–8; $L/s = 13.8$, 15.5, 17.2 for $L = 16, 18, 20$). On that window the slope of F, averaged over one precession period $2\pi/h$ to remove the wiggles, covers about one period, and $\bar{D}$ of $L \leq 20$ already bends towards saturation: even the noise-free $\bar{D}$ of the same L gives only $s = 0.24$–0.31 there, and the Lindblad

values scatter (diffusive 0.17–0.71, jump −0.5–1). $\bar{D}$ also becomes L-dependent before L/s ($\tau \approx 6$ for $L = 22$ against 24). The integrals:

L	12	14	16	18	20	22	24
$\int_0^4 \bar{D}$	27.8	29.0	29.7	30.1	30.4	30.6	30.6
$\int_0^8 \bar{D}$	64.6	70.5	75.8	79.4	82.8	85.0	86.8

$\int_0^4 \bar{D}$ has converged at $L \geq 22$; $\int_0^8 \bar{D}$ still grows by $\approx 2\%$ per two sites at $L = 24$. Consequence: s and the growth law beyond $\tau = 8$ can only come from `single_flip` at $L \geq 24$ (B2) or from a tensor-network single-kick code (C2); a Lindblad run at $L = 22$ (B1) adds the finite-size trend of Γ_e, not s.

A2. How many phases and sites `single_flip` needs. Almost all the spread of $D(\tau)$ is over the phase φ of the kick: a single run differs from $\bar{D}$ by up to 16% ($L = 20$) and 17% ($L = 24$), the site-averaged D varies over phases by up to 16%, the phase-averaged D over sites by only 1.2% and 0.7%. Table 7 gives the error of $\bar{D}$ from n_φ equally spaced phases and n_k equally spaced kicked sites (worst case over τ and over the offsets of the subsample). Four phases are needed, one or two leave $\approx 20\%$; eight reproduce the 16-phase mean to $< 0.01\%$, so the phase dependence has only low harmonics. With four phases, four sites give $\leq 1.1\%$: 16 runs instead of $16 \times L$.

phases × sites	1× all	2× all	4 × 1	4 × 2	4 × 4	4× all	8 × 4
$L = 20$ ($\tau \leq 25$)	26%	21%	3.4%	1.7%	1.1%	0.5%	0.6%
$L = 24$ ($\tau \leq 8$, 4 sites)	28%	21%	1.3%	1.1%	0.6%	–	$< 0.01\%$

Table 7: Worst relative error of $\bar{D}(\tau)$ from a subsample of the 16 phases and of the kicked sites.

A3. How many trajectories. With $\sigma_m(t_e)$ the standard error of $m_\perp$ at the $1/e$ time, the spread of one trajectory is $\sigma_1 = \sigma_m\sqrt{n}$, and a relative error $\sigma_\Gamma/\Gamma_e = e\,\sigma_m$ (Sec. 4) of 2% needs $n = (e\,\sigma_1/0.02)^2$ trajectories (Table 8). σ_1 does not depend on L between 16 and 20. The jump/diffusive variance ratio is ≈ 60–100, so jump trajectories cost $\approx 30\times$ more per error bar even after their first-kick saving: new small-γ runs are diffusive. 56 diffusive trajectories give $\leq 2\%$ at $\gamma \leq 0.02$, 112 give ≈ 1.5–2% at $\gamma = 0.05$.

runs ($L = 16, 18, 20$)	σ_1	n for 2% on Γ_e
diffusive, $\gamma = 0.01, 0.02$	0.035–0.049	23–45
diffusive, $\gamma = 0.05$–0.2	0.056–0.085	58–133
diffusive, $\gamma = 0.5$–10	0.09–0.16	155–450
jump, $\gamma = 0.0005$–0.007	0.25–0.43	1200–3500

Table 8: Spread of one trajectory at t_e and the number of trajectories for a 2% error on Γ_e.

A4. How long the new runs must be. t_e of the planned Lindblad runs, as an upper bound from $1/\Gamma_e(L = 20)$ (Γ_e grows with L) and from Eq. (49) with $\bar{D}$ of the same L, continued past $\tau = 8$ either flat or linearly with slope s (Table 9); the runs stop at $1.5\,t_e$.

run	γ	t_e from $L = 20$	t_e single kick	$t_{\max} = 1.5\,t_e$
B1, $L = 22$	0.005	15.9	14.0–15.4	23.9
B1, $L = 22$	0.01	9.4	8.9–9.0	14.1
B1, $L = 22$	0.02	5.8	5.6	8.7
B1, $L = 22$	0.05	3.2	3.0	4.7
B1′, $L = 24$	0.01	9.4	8.8	14.1
B1′, $L = 24$	0.02	5.8	5.5	8.7

Table 9: Predicted $1/e$ times and run lengths of the planned trajectory runs.

9.2 Step B: the planned runs

Table 10, with the changes that follow from step A (56 trajectories at $\gamma \leq 0.02$ in B1; 4×4 single-flip runs per realization in B3). The new disorder realizations are seeds 2, 3, 4. The Leonardo parts run as Slurm jobs (generators in `data/bashes/`); the obelix parts as an unattended queue that keeps running after logout and skips finished runs. Both are being prepared; none of these runs has been done yet.

	run	details	cost (NH)
B1	scar, diffusive, $L = 22$	$\gamma = 0.005, 0.01, 0.02, 0.05$; 56 trajectories at $\gamma \leq 0.02$, 112 at 0.05; $t_{\max} \approx 24, 14, 9, 5$; half-chain S_A, S_2 every $\Delta t = 1$ ($\gamma \leq 0.01$) or 0.5	≈ 13
B1′	scar, diffusive, $L = 24$	$\gamma = 0.01, 0.02$; 112 trajectories (90 GB); optional, after B1	≈ 47
B2	`single_flip`, longer τ	$L = 24$ to $\tau = 16$ (≈ 4 NH); $L = 26$ to $\tau = 14$ (≈ 17 NH, 51 GB; after the $L = 24$ result); 4 phases $\times$ 4 sites; block entanglement every $\Delta\tau = 1$	≈ 5 (+17)
B3	disorder, seeds 2, 3, 4	`single_flip` $L = 20$, 4×4, $\tau \leq 25$ (≈ 0.3 each); diffusive $L = 20$, $\gamma = 0.01, 0.05, 0.2$, 112 trajectories, $1.5\,t_e$ (≈ 1.9 each); the same at $L = 16$ on obelix	≈ 7
B4	channel study, exact ρ	scar, x-Néel, z-Néel $\times$ σ^z, σ^x, σ^y dephasing, depolarizing, decay, thermal ($\bar{n} = 1$); $L = 12$, $\gamma = 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20$ and $\gamma = 0$; purity, $P(n)$, $\langle n_{xy} \rangle$, $P(\ell)$, $S_{\text{op}}(\rho)$; runs stop at $1.5\,t_e$, Zeno-protected runs ($\sigma^{x,y}$ on the in-plane states, σ^z on z-Néel, large γ) cut at $t = 15$; $L = 8, 10$ on obelix	≈ 128 (232 at the caps)
B5	channel study, trajectories	the same three states, Hermitian channels (σ^z, σ^x, σ^y, depolarizing), diffusive $L = 20$, $\gamma \leq 2$, 56 trajectories at $\gamma \leq 0.02$, 112 above (existing σ^z scar and x-Néel runs not repeated); Zeno cut at $t = 8$; $L = 16$ on obelix	≈ 24 (56 at the caps)
C1	noise on one site	each channel, $\gamma_0 = 1$: exact $L \leq 12$ (obelix), trajectories $L = 16$–20; optional	≈ 6
C2	single kick at $L \approx 100$	MPO/MPS version of `single_flip` (new code), to $\tau \approx 20$; the exact $\gamma = 0$ revival is the go/no-go test; optional	10–30

Table 10: Planned runs. Costs from the job generator (`data/bashes/make_lean_jobs.sh`, bench costs of 19 October): B1–B5 (B2 at $L = 24$) 41 Leonardo jobs, ≈ 176 NH $\approx 19\,700$ core-hours expected, 314 NH if every run reaches its time cap; with B2 at $L = 26$: ≈ 193 NH. The obelix queue (`obelix/make_queue.py`, 530 runs: B4 exact at $L = 8, 10$ for every state and channel, $\gamma = 0$ and 0.005–20; B3 at $L = 16$; B5 diffusive at $L = 16$) takes ≈ 52 hours. For comparison, the small-γ batch of 19–20 October used $\approx 24\,100$ core-hours.

Order: B4 and the obelix parts of B3 and B5 first (they need the new code of Secs. 3.8–3.9),

then B1 and B2 at $L = 24$, then B3 on Leonardo; B1′, B2 at $L = 26$ or C2 are decided from those results.

10 Entanglement of eigenstates and Liouvillian modes

Sanity check 3 of the project plan: the bulk eigenstates should form a thermal dome and the scars should be low-entanglement outliers. Same random-c realization as everywhere else (seed 1), periodic chain; the cut is sites $0, \dots, L_A - 1$, $L_A = \lfloor L/2 \rfloor$.

Eigenstates of H. All 2^L eigenstates of H by dense diagonalization (`entanglement`, $L = 6$–12), with the half-chain entropy S_A and the tower weight $\sum_n |\langle \Phi_n | E \rangle|^2$ (Fig. 15). Exactly $L + 1$ eigenstates have tower weight above 1/2 (it is 1 to 10^{-6}), at $E = h(L/2 - n)$: the scars. Their entropies are those of the Dicke states, $S = -\sum_k p_k \ln p_k$ with $p_k = \binom{L_A}{k}\binom{L-L_A}{n-k}/\binom{L}{n}$, to 5×10^{-13}. The two edge scars are product states ($S = 0$); the middle one has $S = 1.02$ ($L = 6$) to 1.32 ($L = 12$), growing roughly as $\frac{1}{2} \ln L$. The bulk forms a dome whose middle approaches the Page value $S_{\text{Page}} = L_A \ln 2 - 2^{2L_A - L - 1}$: the mean S_A of the bulk states with $|E| < 1$ is 0.90, 0.92, 0.92, 0.95, 0.96, 0.97, 0.97 of it for $L = 6, \dots, 12$. Low-entanglement bulk states exist only at the edges of the spectrum: at $|E| < 1$ the least entangled bulk state has $S_A = 3.41$ at $L = 12$, against 1.32 for the scar there.

Operator entanglement of $|n\rangle\langle m|$. The operator Schmidt decomposition of $|n\rangle\langle m|$ is the product of the Schmidt decompositions of $|n\rangle$ and $|m\rangle$, so its operator entanglement is exactly $S_n + S_m$ (checked by SVD for random states, 2×10^{-15}); no further diagonalization is needed (Fig. 16). Three families separate: bulk–bulk pairs up to twice the Page value, scar–bulk pairs in between, and scar–scar pairs, the tower coherences, below 2.6 at $L = 12$.

Liouvillian modes. Right eigenmatrices X of $\mathcal{L}$ by dense diagonalization (`lindblad_entanglement`; $\gamma = 0$ for $L = 4$–7, $\gamma = 0.01$ for $L = 4$–6). From each X: $\rho_L = XX^\dagger/\operatorname{Tr}$, $\rho_R = X^\dagger X/\operatorname{Tr}$, their energies $E = \operatorname{Tr}(\rho H)$, the operator entanglement S_{op} (at most $2L_A \ln 2$; $2S$ for a pure state) and the tower weight $\|P_{\text{tower}} X P_{\text{tower}}\|/\|X\|$. $\mathcal{L}$ has degenerate eigenspaces (the kernel, the edge pairs $|\Phi_0\rangle\langle\Phi_n|$, $|\Phi_{L-n}\rangle\langle\Phi_L|$, and at $\gamma = 0$ every $|n\rangle\langle n|$ and every tower spacing); there the basis is the one that diagonalizes $\operatorname{Tr}(X_i^\dagger H X_j)$ against $\operatorname{Tr}(X_i^\dagger X_j)$, which splits each edge pair exactly. Checks: at $\gamma = 0$ the modes are the $|n\rangle\langle m|$, ρ_L is pure and $S_L = 2S_n$, equal to the eigenstate values to 7×10^{-12}; the edge modes give the Dicke-state E and S to 10^{-13}; since $\mathcal{L}$ is real, the (E_R, S_R) and (E_L, S_L) sets coincide.

At $\gamma = 0$ the $(L+1)^2 = 25, 36, 49, 64$ tower coherences have tower weight 1. At $\gamma = 0.01$, 22, 32, 44 modes keep tower weight above 0.9 ($L = 4, 5, 6$): the dephasing mixes a few mid-tower coherences with the bulk (as in the earlier spectrum, App. A). These modes stay low-entanglement outliers (Fig. 17): $S_L \leq 1.74$, 1.81, 2.05 for $L = 4, 5, 6$, against the maximum 2.77, 2.77, 4.16 and the median over all modes 1.39, 1.82, 2.60; from $\gamma = 0$ to 0.01 their largest S_L changes by less than 0.02. Points at $S_L = 0$ are the modes with $\rho_L = |\Phi_0\rangle\langle\Phi_0|$ or $|\Phi_L\rangle\langle\Phi_L|$; at $\gamma > 0$ these include the 2^{L+1} half-dark modes $|\Phi_0\rangle\langle\psi|$, $|\Phi_L\rangle\langle\psi|$, which have tower weight ≈ 0 but the same ρ_L. Only $L \leq 7$ is accessible this way (4^L-dimensional dense diagonalization; $L = 7$ takes 18 minutes and 4.4 GB on the laptop).

11 Tensor-network and Krylov (CLIK) results

TDVP of ρ from the scar state (previous coefficients, $\gamma = 0.01$, $t \leq 10$, $\delta t = 0.1$). The MPS of $\rho(t)$ reaches its maximal bond dimension by $t \approx 0.5$. Against the exact evolution, the error of the magnetization is 10^{-2} at $\chi = 32$ and 8×10^{-5} at $\chi = 64$ for $L = 6$ (full $\chi = 4^3 = 64$), and 2×10^{-2} at $\chi = 64$ for $L = 8$ (full 256). The dissipated scar state needs nearly the full bond

dimension: the weight that the dephasing moves out of the tower, at rate $\approx \gamma(L-1)$ (Eq. (10)), is thermalized by H_I, and that part has large operator entanglement. B4 measures $S_{\mathrm{op}}(\rho(t))$ directly.

CLIK (current coefficients, random c, $\gamma = 0.01$, $\delta t = 0.5$, $K = 4$ product terms per initial state; errors against the exact spectrum of $\mathcal{L}$).

- **Checks.** The Julia $\mathcal{L}$ (MPO, contracted) has the Rust ED spectrum to 5×10^{-12} ($L = 4, 5, 6$, ring and open chain). At the full bond dimension $4^{L/2}$ the MPS backend reproduces the dense Krylov matrices to 10^{-12} ($L = 4$) and 10^{-13} ($L = 6$), but only with the global subspace expansion; without it two-site TDVP misses the range-4 terms while the bonds are small (10^{-2} error from the first step).

- **The complex contour does not help here.** From the ED, $\alpha_{\max} = 0.0020$, 0.0031, 0.0027 (ring, $L = 4, 5, 6$) and 0.0037 (open, $L = 6$), set by bulk modes with $\mathrm{Im}\,\lambda \approx$ 15–23, not by the slow modes. The boost of slow modes is therefore negligible and CLIK works as a real-time Krylov method, converging after $T \approx 3$–$5/\Delta \approx 200$–300 ($\Delta \approx 0.015$–0.018).

- **Ring, random initial states, dense,** $T = 400$**:** $\lambda = 0$ to 10^{-15}; the edge mode $-0.02 + i$ to 3×10^{-14}, 10^{-10}, 10^{-9} ($L = 4, 5, 6$); the slowest bulk mode to 8×10^{-11}, 3×10^{-8}, 3×10^{-7}; some real bulk modes never converge (10^{-3}–10^{-2}).

- **Open chain,** $L = 6$**, scar initial states** (mixtures of spin coherent states $|\mathbf{n}\rangle\langle\mathbf{n}|^{\otimes L}$, all inside the tower) against random ones, dense, $T = 200$: the tower modes converge 2–4 orders of magnitude better ($\lambda = 0$: 8×10^{-11} against 2×10^{-9}; $-0.02 + i$: 3×10^{-9} against 10^{-5}; $-0.04 + 2i$: 3×10^{-7} against 5×10^{-3}).

- **Truncation** (open chain, scar initial states, $\chi = 16$ and 32 of the full 64): only $\lambda = 0$ (4×10^{-8}, 2×10^{-7}) and the first edge mode (9×10^{-6}, 2×10^{-5}) converge; every other tower mode stays at $\approx 10^{-3}$, and the trace of the traceless state leaks to 3×10^{-3} and 7×10^{-3}. This is not a time-step error: at $t \le 10$ and $\chi = 32$ the Krylov matrices differ from the dense ones by 3×10^{-3} for internal steps 0.25 and 0.05 alike. As for TDVP: $\rho(t)$ from scar states needs nearly the full bond dimension at $L = 6$.

Forward/backward evolution (`julia/forward_backward.jl`, current coefficients, random c). Eq. (8) in Pauli coordinates: ρ is evolved with G, the observable coordinates $o_P = \mathrm{Tr}(AP)/2^L$ with G^{T}, and $\langle A\rangle_{t_1+t_2} = o(t_2)\cdot r(t_1)/\mathrm{Tr}\,\rho(t_1)$. G^{T} is built as an MPO from the same terms: the Hamiltonian part with $H \to -H$, the dissipator diagonal unchanged, the feed of decay and thermal noise transposed ($\mathbb{1} \to \sigma^z$); checked against the transpose of the contracted G to 5×10^{-15} at $L = 6$. The same run also gives the forward-only and backward-only values at the same χ, and the spread of $\langle A\rangle_t$ over all splits $t_1 + t_2 = t$, a truncation-error estimate.

- **Time stepping.** Two-site TDVP is wrong while the bond dimension is small: one step $\delta t = 0.1$ from $|-\rangle^{\otimes6}$ is off by 10^{-2} in norm, with or without global Krylov expansion, scaling as δt^2 (the range-4 terms of H leave the two-site tangent space). A Taylor series of $e^{G\tau}$ with exact MPO applications has no such error (6×10^{-8} for the same step, χ stays 1 as it must at $\gamma = 0$) but costs ≈ 25 s per substep at $\chi = 64$. The code uses Taylor while $\chi < 16$ and TDVP above.

- **Validation at** $L = 6$ (full $\chi = 64$, $\delta t = 0.1$, $t \le 5$, against the exact Rust evolution). Scar at $\gamma = 0$ (perfect revivals): largest deviation of m_x, m_y 2.5×10^{-6} (forward only), 1.3×10^{-6} (meet in the middle), 8×10^{-9} (backward only); all six magnetizations below 3×10^{-6}. Scar, σ^z dephasing, $\gamma = 0.1$: below 3×10^{-7} (forward, middle) and 5×10^{-9} (backward). Other channels at $\gamma = 0.1$: scar under σ^x dephasing 9×10^{-6} and under decay 2×10^{-6} (forward, middle; backward $\le 7\times10^{-9}$); x-Néel (s_x, s_y) under σ^z dephasing and decay, and z-Néel (s_z) under σ^z, σ^x dephasing and decay: $\le 6\times10^{-8}$ for all three. The backward half is the

more accurate one from the scar state, because the forward state stays at small χ (where TDVP is not used, but the Taylor steps are compressed with a cutoff) for longer.

- **Cost.** At $L = 6$ and full χ one run ($t \leq 5$, one or two observables) takes 150–260 s on the laptop (8 threads); with all six observables 370 s ($\gamma = 0.1$) and 1070 s ($\gamma = 0$). Dense vectors are far cheaper at this size; the method is meant for $L \gtrsim 14$, where it has not been run yet.

12 Summary and open points

- The scar transverse magnetization decays at an L-independent rate once $L - 1 \gg \Gamma_\infty/\gamma$. Its thermodynamic rate vanishes as $\gamma \to 0$, roughly as $\gamma^{1/2}$ for $0.02 \leq \gamma \leq 0.2$, and tends to 2γ at large γ. The thermal x-Néel signal decays at the closed-system rate ≈ 4.8 for $\gamma \leq 0.1$, so the two differ by a factor 37 at $\gamma = 0.01$. The z-Néel state is Zeno-protected and not a fair reference.

- At fixed L and $\gamma \to 0$, $\Gamma_e/\gamma \to L - 1$, the loss rate of the tower weight: the local signal then behaves like a global one.

- Extrapolating only the large sizes ($L = 14$–20) gives estimates of Γ_∞ down to $\gamma = 0.0005$ ($\Gamma_\infty/\gamma \approx 18$–$34$ for $\gamma < 0.01$, not converged), consistent with $\Gamma_\infty \propto \gamma^{0.58}$–$\gamma^{0.66}$ over $0.0005 \leq \gamma \leq 0.05$ and with the collapse.

- A single kick evolved with H predicts the scar decay at every $L \leq 20$ and $\gamma \leq 0.05$ to within $\approx 5\%$ without parameters. The small-γ lifetime is therefore the integral of the single-kick damage.

- Entanglement (Sec. 10): the $L+1$ scars are the Dicke states, with $S \leq 1.32$ at $L = 12$ against a bulk dome that reaches 97% of the Page value in the middle of the spectrum. Among the Liouvillian modes ($L \leq 6$, $\gamma = 0.01$) the tower coherences stay low-entanglement outliers, almost unchanged from $\gamma = 0$.

- Next (Sec. 9): the existing data confirm the single-kick formula for the whole $m_\perp(t)$ and show that the growth of $\bar{D}$ must come from single-kick runs at $L \geq 24$; the planned runs (≈ 40 NH, 57 with $L = 26$) add $L = 22$ rates, longer $\bar{D}(\tau)$, three more disorder realizations, operator observables of ρ and five other noise channels.

- Theory (Sec. 2): at fixed L and $\gamma \to 0$ first-order perturbation theory gives the cap $\Gamma = \gamma(L - 1 + 2/L)$; in the thermodynamic limit the single-kick damage growth $\bar{D}_\infty \propto \tau^\beta$ gives $\Gamma_\infty \propto \gamma^{1/(1+\beta)}$ and the collapse exponent $\nu = \beta/(1 + \beta)$ (ballistic: $\gamma^{1/2}$, $\nu = 1/2$; diffusive: $\gamma^{2/3}$, $\nu = 1/3$).

- Tensor networks (Sec. 11): TDVP and truncated CLIK need nearly the full bond dimension for $\rho(t)$ from the scar state, because the weight that leaves the tower thermalizes and carries operator entanglement.

- Open: the growth law of $\bar{D}_\infty(\tau)$ (ballistic or diffusive) at $\tau > 8$, which fixes the exponent of Γ_∞ as $\gamma \to 0$; one disorder realization per L; the $\gamma = 0.01$–0.02 extrapolation, which is the least converged.

13 Data computed so far

Everything the current analysis uses is in `data/used_data/` (not tracked by git; 505 files); older and superseded data are in `data/non_used_data/`. All runs with the current coefficients use the same random-c realization per L (seed 1), stored as `states_realization_L<L>.csv` for $L = 4, \ldots, 12, 14, \ldots, 24$. Standard γ grid: $\gamma \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10\}$. Small-$\gamma$ grid:

$\{0.0005, 0.0007, 0.001, 0.0014, 0.002, 0.003, 0.004, 0.005, 0.007\}$. Methods as in Sec. 3; cluster runs on Leonardo (Cineca), the rest on the laptop.

13.1 Time evolution, current coefficients

Files `states_<state>_L<L>_g<γ>.csv` (exact), `states_traj_<state>_L<L>_g<γ>.csv` (diffusive) and `states_traj_<state>_L<L>_g<γ>_jump_s<N>.csv` (jump, one file per noise seed N); columns t, $m_{x,y,z}$, $s_{x,y,z}$ (and their standard errors for trajectories). Runs stop once the signal stays below $\max(10^{-3}, 3\sigma)$ for 10 time units; $\gamma = 0$ runs to $t = 50$; the small-γ runs stop at $t_{\max} = 3/(\gamma(L-1))$.

state	method	L	γ	trajectories	batch
scar	exact ρ	$4, \ldots, 12$	standard grid	–	Oct 15
scar	exact ρ	14	standard grid; $\gamma \leq 0.02$ cut at 24 h ($t \approx 25$–32)	–	Oct 16
scar	exact ρ	$8, 10$	small-γ grid	–	Oct 19–20
scar	exact ρ	12	0.002–0.007 (5 values)	–	Oct 19–20
scar	diffusive	$12, 14, \ldots, 20$	standard grid	224 (112 at $L = 20$)	Oct 16
scar	diffusive	$12, 14$	$0.002, 0.005$ (cross-check)	448	Oct 19–20
scar	jump	$12, 14, 16$	small-γ grid	448	Oct 19–20
scar	jump	$18, 20$	$0.0005, 0.001$; 0.002–0.007	224; 448	Oct 19–20
x-Néel	exact ρ	$4, 6, \ldots, 14$	standard grid ($L = 14$, $\gamma \leq 0.02$ cut at 24 h)	–	Oct 16
x-Néel	diffusive	$12, 14, \ldots, 20$	standard grid	224 (112 at $L = 20$)	Oct 16
z-Néel	exact ρ	$4, 6, \ldots, 12$	standard grid	–	Oct 15

Table 11: Time-evolution runs with the current coefficients (random c). Jump runs use output step 0.25, all others $\min(0.1, 0.05/\gamma)$. At $\gamma = 0$ the trajectory runs are one deterministic trajectory, to $t = 50$ ($t = 20$ at $L = 18, 20$). Batches: Leonardo, October 2026. Exact runs: 133 (scar), 66 (x-Néel), 55 (z-Néel) files; trajectories: 59 diffusive and 46 jump files (scar), 55 diffusive (x-Néel). One run of the small-γ batch ended early by a node failure ($L = 18$, $\gamma = 0.001$, jump: $t = 125.8$ of 176.5, past t_e).

13.2 Single-kick runs

L	phases	kicked sites	$\tau_{\max}$	output step	file
$12, 14, 16, 18, 20$	16	all L	25	0.1	`single_flip_L<L>.csv` (30–81 MB)
22	16	$0, 5, 11, 16$	8	0.1	`single_flip_L22_sites0-5-11-16.csv`
24	16	$0, 6, 12, 18$	8	0.1	`single_flip_L24_sites0-6-12-18.csv`

Table 12: `single_flip` runs (Leonardo, Oct 19–20): one row per (phase, kicked site, τ) with D and, per site, $\mathrm{Re}\, c_j$, $\mathrm{Im}\, c_j$, $\langle \sigma_j^z \rangle$.

13.3 Spectra, entanglement and Lindbladian eigenvalues (laptop)

- **Eigenstates of H:** `entanglement_L<L>.csv`, $L = 6, \ldots, 12$ (random c): energy, half-chain entropy and tower weight of all 2^L eigenstates (2026-10-22).
- **Liouvillian modes:** `lindblad_entanglement_L<L>_g<γ>.csv`, $\gamma = 0$ for $L = 4, \ldots, 7$ and $\gamma = 0.01$ for $L = 4, 5, 6$: every eigenvalue of $\mathcal{L}$ with $E_{L,R}$, $S_{L,R}$ and the tower weight (2026-10-22).
- **Open chain, full ED of $\mathcal{L}$:** `spectrum_open_L<L>.csv`, $L = 4, 5, 6$, $\gamma = 0.01$ (reference for CLIK).

- **CLIK** (Lindbladian eigenvalues from time-evolved states, `julia/clik.jl`, $\gamma = 0.01$, 2026-10-23; results in Sec. 11): dense backend for the ring at $L = 4, 5, 6$ (random initial states) and the open chain at $L = 6$ (random and scar initial states); MPS backend for the ring at $L = 6$, $\chi = 64$, and the open chain at $L = 6$ with scar initial states, $\chi = 16, 32$; full ED references `clik_ed_L<L>_g0.01[_open].csv`; errors in `clik_errors.csv`. The Hamiltonians are exported as `data/hamiltonian[_open]_L<L>.csv` ($L = 4, \ldots, 10$; open up to 12).

13.4 Derived tables

Produced by the scripts from the runs above, in `data/used_data/`:

- `states_rates.csv`: Γ_e and Γ_{tail} of every run (`fit_states.py`);
- `states_inverse_L.csv`: Γ_∞ from $A + B/L$ on $L \geq 8$ (`fit_states_inverse_L.py`);
- `states_inverse_L_large.csv`: the scar fit on $L \geq 14$;
- `small_gamma_rates.csv`, `small_gamma_checks.csv`, `small_gamma_collapse.csv`: scar Γ_e with statistical errors, jump against exact and diffusive, ν and its range (`fit_small_gamma.py`);
- `single_kick_damage.csv`, `single_kick_rates.csv`: $\bar{D}(\tau)$ and measured against predicted Γ_e (`single_kick.py`);
- job costs in `data/non_used_data/logs/` (Oct 16 batch, jump bench, small-γ batch, each with `jobs_summary.csv`); machine benchmarks in `data/*_benchmarks.md` and `data/non_used_data/bench_tests/`.

13.5 Analysis outputs of step A

From `plot_scripts/lean_plan_analysis.py` (Sec. 9), in `data/used_data/`:

- `lean_A1_damage.png`: $F(t)$ against $\int \bar{D}$ (Fig. 14);
- `lean_A2_subsampling.csv`: error of $\bar{D}$ from subsamples of phases and sites;
- `lean_A3_spread.csv`: spread of one trajectory at t_e and the trajectory counts for 2%;
- `lean_A4_t_e.csv`: predicted t_e and $t_{\max}$ of the planned runs.

The step-B runs of Table 10 have not been done yet.

13.6 Older data (`data/non_used_data/`)

- **Previous coefficients** (`previous_coefficients/`, laptop, 13–14 October 2026; Appendix A): exact evolution of the scar, uniform c, $\gamma = 0$ and 0.01, $L = 4, \ldots, 10$, $t \leq 100$ (`evolution_L<L>.csv`), and from random product states (`evolution_random_L<L>.csv`); diffusive trajectories $L = 4, \ldots, 8$, $\gamma = 0.01$ (2000 trajectories; 20000 at $L = 4, 5$); TDVP of ρ as an MPS, $L = 6$ ($\chi = 8$–64) and $L = 8$ ($\chi = 16$–128); cone initial states $L = 4, \ldots, 8$ (`disorder_L<L>.csv`); probes P1, P2, P3 and the free-spin check, $L = 4, \ldots, 8$, $\gamma = 0, 0.01$, and the γ scan at $L = 6$ ($\gamma = 0.002$–1, 3, 6); Liouvillian ED at $\gamma = 0.01$, $L = 4, 5, 6$ (`spectrum_L<L>.csv`); first-pass envelope runs, $L \leq 8$, $\gamma = 0.01$–0.2 (`first_pass/`).
- **Current coefficients, uniform** c (`uniform_c_2026-10-24/`): exact evolution of the scar, $\gamma = 0$ and 0.01, $L = 4, \ldots, 10$, $t \leq 100$; not used.
- **Superseded:** the first cluster batch (`states_batch_2026-10-15/`, identical run files are in `used_data`), old analysis outputs (`states2_leftovers/`), and the archives received from Leonardo (`archives/`).

A Earlier results with the previous coefficients ($L \leq 10$)

The results of this appendix (14 October 2026) were obtained with $h = 0.5$, $a = (0.73, 0.41, 0.29)$, $b = (0.58, -0.37, 0.66)$, and only c_{xz}, c_{zy}, c_{yx} nonzero with baseline values $0.31, -0.44, 0.27$ (random factors in $[0.5, 1.5]$ where stated), with exact evolution and full Liouvillian ED for $L \leq 10$, for the scar state and the probes below (the tilted cone initial states studied then are left out). Rates are fits of $A_0 e^{-\Gamma t}$ with A_0 free in fixed windows, not the $1/e$ rates of the main text (Sec. 4).

The tables of this appendix keep those numbers. Figures 18–20 are redrawn from the runs with the current coefficients (random c, Secs. 6, 7) and show Γ_e: the same quantities where the current runs have them, the x- and z-Néel states in place of the probes P2, P3, which were not rerun.

A.1 Scar oscillations and their decay

Decay against L. The amplitude $m_\perp$ decays faster for larger L (Fig. 18). For the translation-invariant model, fitting $e^{-\Gamma t}$ while $m_\perp > 0.05$ gives the rates in Table 13, well described by $\Gamma \approx 0.63\,\gamma L^{1.13}$. Breaking translation with random c changes them by less than 2%. With the main-text results this growth is identified as the finite-size cap of Sec. 7.

L	4	5	6	7	8	9	10
Γ/γ, uniform c	3.06	3.80	4.66	5.75	6.68	7.61	8.39
Γ/γ, random c	3.06	3.89	4.68	5.81	6.69	–	–

Table 13: Scar decay rate at $\gamma = 0.01$, fit window $m_\perp > 0.05$ (previous coefficients).

A.2 Scar versus non-scar probes

Three probes, evolved with the same Lindbladian (random c, $L = 4, \ldots, 8$, $\gamma = 0$ and 0.01):

- **P1, scar:** $m_\perp(t)$ from $\rho_0 = |-\rangle\langle -|^{\otimes L}$.
- **P2, energy:** $C(t) = \mathrm{Tr}[\tilde{H} e^{\mathcal{L}^\dagger t}(\tilde{H})]/\,\mathrm{Tr}\,\tilde{H}^2$ with $\tilde{H} = H - hS^z_{\mathrm{tot}} P_{\mathrm{tower}} = (1 - P_{\mathrm{tower}})H(1 - P_{\mathrm{tower}})$, conserved at $\gamma = 0$ and without tower component.
- **P3, thermal:** $|C^+(t)|$ for $O^+ = \sum_j (-1)^j \sigma^+_j$ ($k = \pi$, even L), built from separate evolutions of O^x_π and O^y_π so that the precession in h drops out. At $\gamma = 0$ it thermalizes by $t \approx 1$ and then sits at a finite-size floor ($0.17, 0.05, 0.015$ for $L = 4, 6, 8$). Its dephasing rate is $\Delta\Gamma$, the slope of $-\ln(|C^+_\gamma|/|C^+_0|)$ on a common short window. $\mathrm{Tr}(O_k H) = 0$ exactly for $k \neq 0$ (the single-site σ^x content of H is uniform).

Scaling with L. Table 14 and Fig. 20. The scar rate grows close to linearly in L; the non-scar rates stay near the single-spin value 2γ with a weak (energy) or no (thermal) L-dependence.

probe (Γ/γ)	$L = 4$	5	6	7	8	power law
P1 scar, window $t \in [5, 60]$	3.22	4.01	4.69	5.67	6.51	$L^{1.01\pm0.03}$
P1 scar, window $t \in [5, m_\perp = 10^{-3}]$	2.61	2.94	3.28	3.87	4.77	$L^{0.84\pm0.12}$
P2 energy $\tilde{H}$	1.83	1.84	2.00	2.22	2.32	$L^{0.37\pm0.07}$
P3 thermal $\Delta\Gamma$, $t \in [0, 0.5]$	1.43	–	1.59	–	1.61	$L^{0.18\pm0.06}$
P3 thermal $\Delta\Gamma$, $t \in [0, 1]$	2.25	–	2.25	–	1.85	$L^{-0.27\pm0.19}$

Table 14: Decay rates at $\gamma = 0.01$ (random c, previous coefficients) and power-law fits $\Gamma = aL^b$.

Caveats.

- The scar decay is multi-exponential (Fig. 19, left): mid-tower coherences ($\sim \gamma L$) dominate early, the exact 2γ edge modes late. Γ_{scar} and its L-exponent depend on the fit window.
- $L = 4, \ldots, 8$ is a factor of two in L: power law and exponential fit about equally well; the exponents are descriptive.
- For P3, a window that ends at twice the noise floor lengthens with L and produces a spurious growth of $\Delta\Gamma$ $(1.43, 2.25, 4.39)$; only common windows are meaningful.
- The predicted initial rate of P2, $2\gamma \sum_P n_{xy} h_P^2 / \sum_P h_P^2$, is 2.7–2.9 γ; the long-time decay is slower (1.8–2.3 γ).

A.3 Liouvillian spectrum

Full ED at $\gamma = 0.01$ for $L = 4, 5, 6$ (Fig. 21). Each right eigenvector R is split into tower ($\|P_{\text{tower}} R P_{\text{tower}}\|$), mixed and bulk weights. Findings:

- three zero modes (Sec. 5), and $\operatorname{Re}\lambda \geq -2\gamma L$ saturated by $|\Phi_0\rangle\langle\Phi_L|$;
- the bulk is centred at $-\operatorname{Re}\lambda \approx \gamma L$ (the mean dephasing rate of a Pauli string);
- the slowest tower modes sit at exactly 2γ (edge coherences); the next ones at 4γ for $L = 5, 6$ (2.94γ at $L = 4$);
- the slowest bulk mode is at 1.67γ, 1.66γ, 1.88γ for $L = 4, 5, 6$, comparable to the P2 rate;
- modes with tower weight > 0.9: 22 of 25, 32 of 36, 38 of 49: at $\gamma = 0.01$ some tower coherences already hybridize with the bulk.

Checks: at $\gamma = 0$ the dense spectrum equals $-i(E_m - E_n)$ to 5×10^{-14} ($L = 4, 5$); $\sum \lambda = \operatorname{Tr}\mathcal{L} = -\gamma L 4^L$; $L + 1$ eigenstates of H in the tower at energies $h(L/2 - n)$.

A.4 Dependence on γ at fixed $L = 6$

Eleven values $\gamma = 0.002, \ldots, 6$, rates fitted on a window fixed in $\gamma t \in [0.05, 0.6]$ (the window $[5, 60]$ at $\gamma = 0.01$).

γ	0.002	0.005	0.01	0.02	0.05	0.1	0.2	0.5	1	3	6
P1 scar	4.80	4.78	4.69	4.61	4.59	5.67	3.75	2.47	2.14	2.02	2.00
P2 energy	2.44	2.42	2.39	2.36	2.26	2.09	1.93	1.62	1.42	1.29	1.27
P3 $\Delta\Gamma$	1.60	1.60	1.60	1.60	1.60	1.61	1.62	1.66	1.72	1.88	2.28

Table 15: Γ/γ at $L = 6$ (random c, previous coefficients).

Three regimes for the scar (Table 15): weak dephasing, $\gamma \leq 0.05$, with $\Gamma \propto \gamma^{0.984 \pm 0.002}$ at $\approx 4.7\gamma$ (the cap $\approx L - 1$ of the main text); a crossover around $\gamma \approx 0.1$, where $\gamma L \approx h$, with a maximum 5.7γ; strong dephasing, $\gamma \gtrsim 1$, with $\Gamma \to 2\gamma$. The non-scar rates are also linear in γ for weak dephasing ($\gamma^{0.977}$ for P2, $\gamma^{1.001}$ for P3). At strong dephasing the energy probe slows down (1.27γ at $\gamma = 6$).

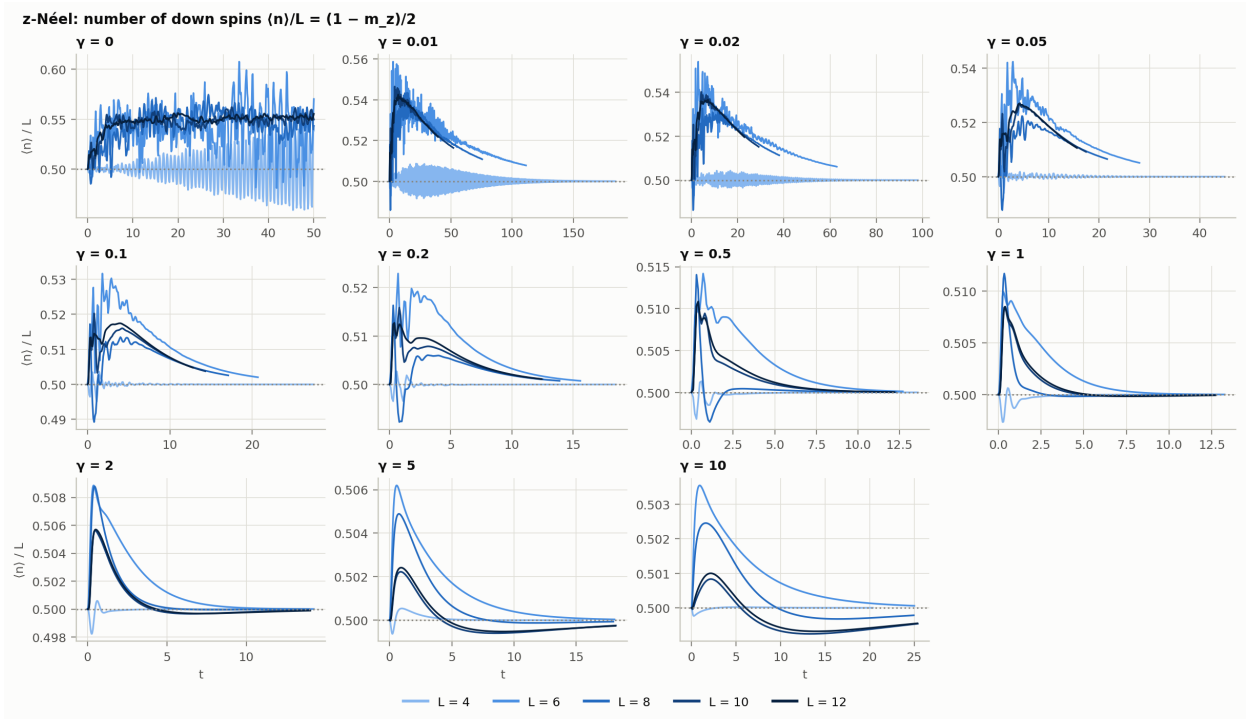


Figure 5: z-Néel: $\langle n\rangle/L = (1 - m_z)/2$; the runs stop on $|s_z|$, so $\langle n\rangle$ is not fully relaxed at small γ (`plot_states.py`).

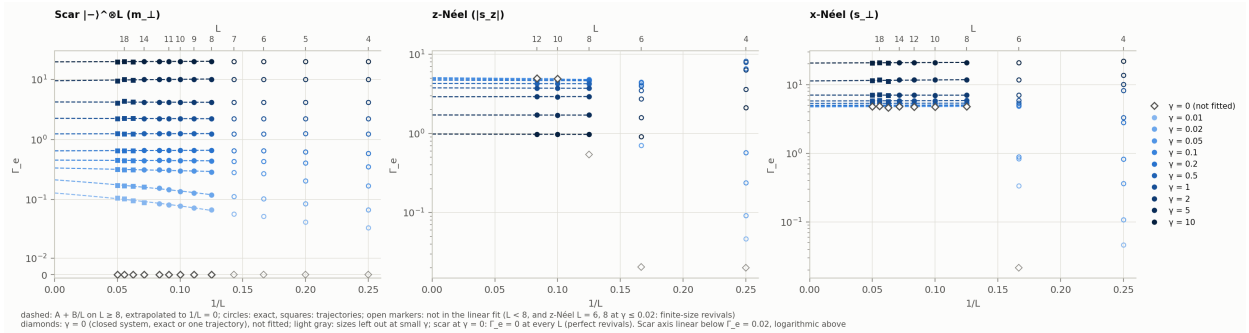


Figure 6: Γ_e against $1/L$ with the fits $A + B/L$ on $L \geq 8$; circles exact, squares trajectories; diamonds $\gamma = 0$, not fitted (`fit_states_inverse_L.py`).

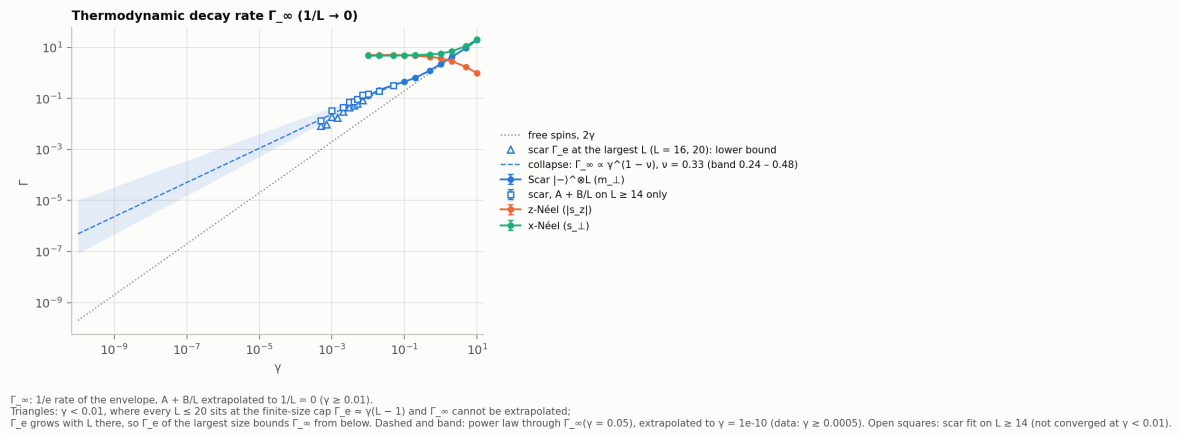


Figure 7: Γ_∞ against γ for the three states (log-log, $\gamma > 0$ only). Scar at $\gamma < 0.01$ (Sec. 7): every $L \leq 20$ sits at the finite-size cap, so Γ_∞ is not extrapolated; triangles are Γ_e of the largest size ($L = 20$; 16 at $\gamma = 0.0007, 0.0014$), lower bounds on Γ_∞ since Γ_e grows with L there. Open squares: $A + B/L$ on $L \geq 14$ only (Table 6; not converged at $\gamma < 0.01$). Dashed line and band: $\Gamma_\infty \propto \gamma^{1-\nu}$ through $\Gamma_\infty(0.05)$ with the collapse exponent $\nu = 0.33$ (0.24–0.48), extrapolated to $\gamma = 10^{-10}$ (data from $\gamma = 5 \times 10^{-4}$; free-spin 2γ over the same range) (`fit_states_inverse_L.py`, `fit_small_gamma.py`).

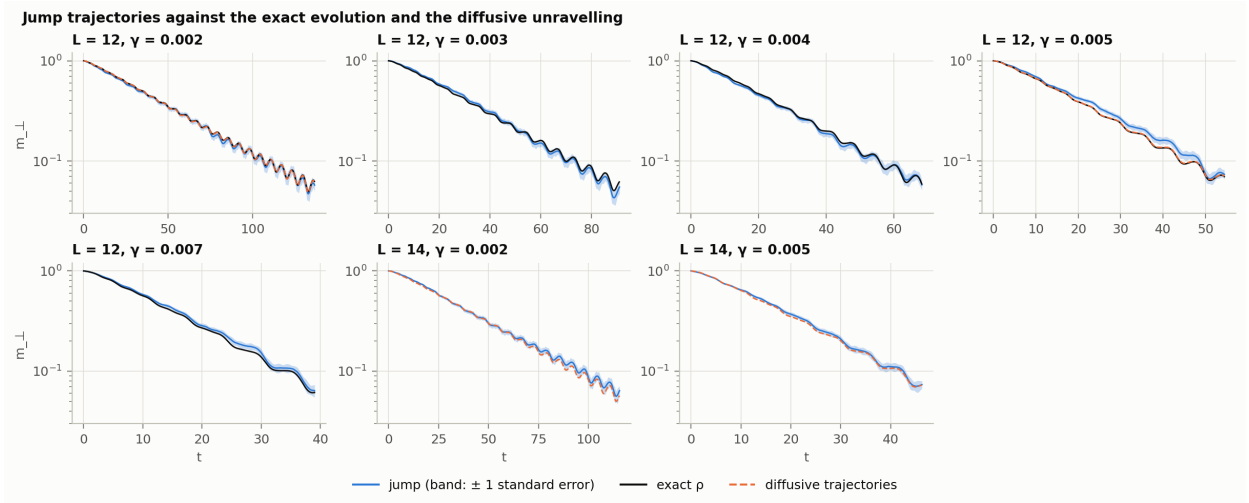


Figure 8: $m_\perp$ of jump trajectories (band: ± 1 standard error) against exact and diffusive runs (`fit_small_gamma.py`).

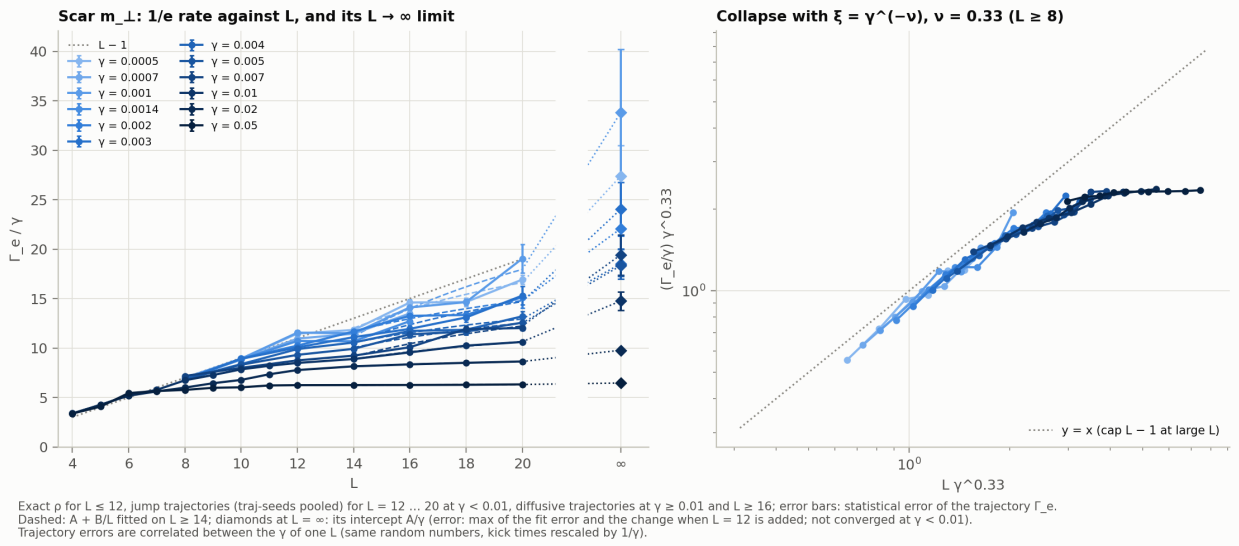


Figure 9: Left: Γ_e/γ against L, one line per γ; dotted: $L-1$; dashed: $A+B/L$ fitted on $L \geq 14$, and diamonds at $L=\infty$: its intercept A/γ (Table 6; not converged at $\gamma < 0.01$). Right: collapse with $\nu = 0.33$ (`fit_small_gamma.py`).

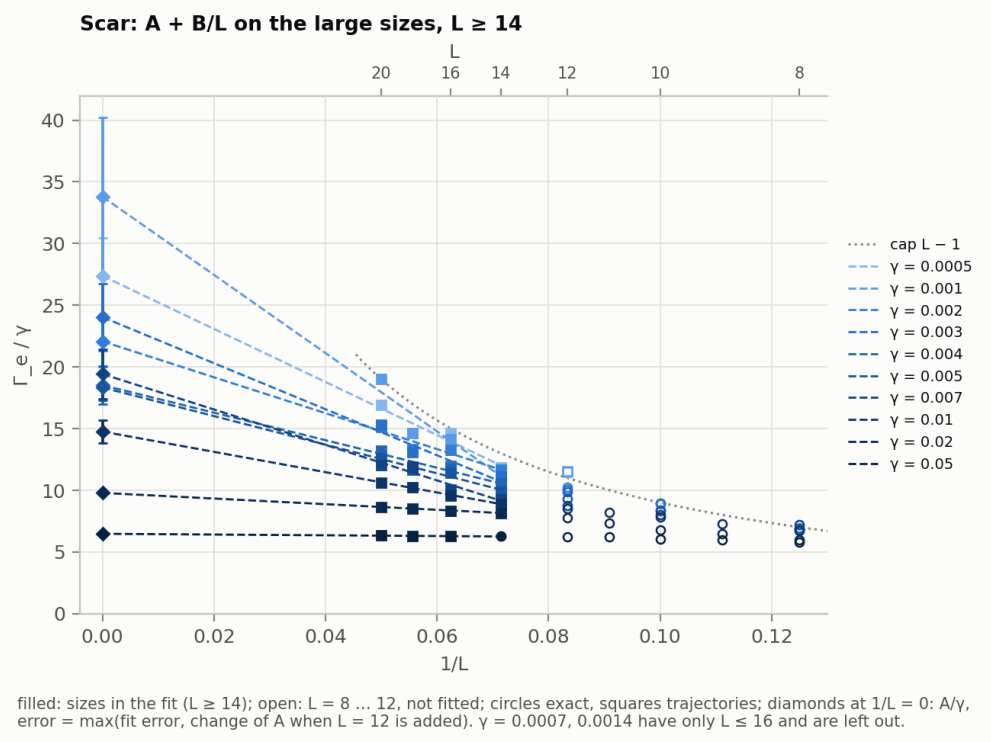


Figure 10: Scar Γ_e/γ against $1/L$ for $\gamma \leq 0.05$, with $A+B/L$ fitted on $L \geq 14$ (filled) and extrapolated to $1/L = 0$ (diamonds); open symbols: $L \leq 12$, not fitted; dotted: the cap $L-1$ (`fit_small_gamma.py`).

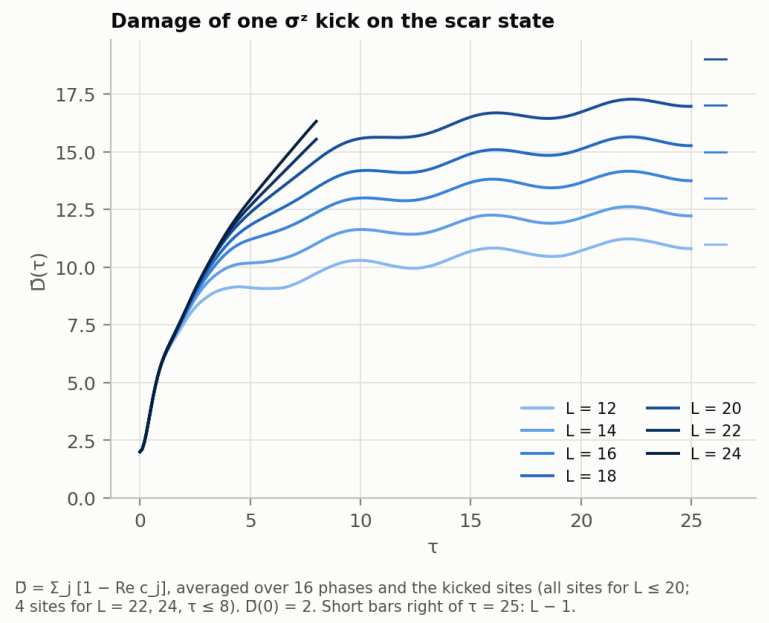


Figure 11: Damage $\bar{D}(\tau)$ of one kick; bars right of $\tau = 25$: $L - 1$ (`single_kick.py`).

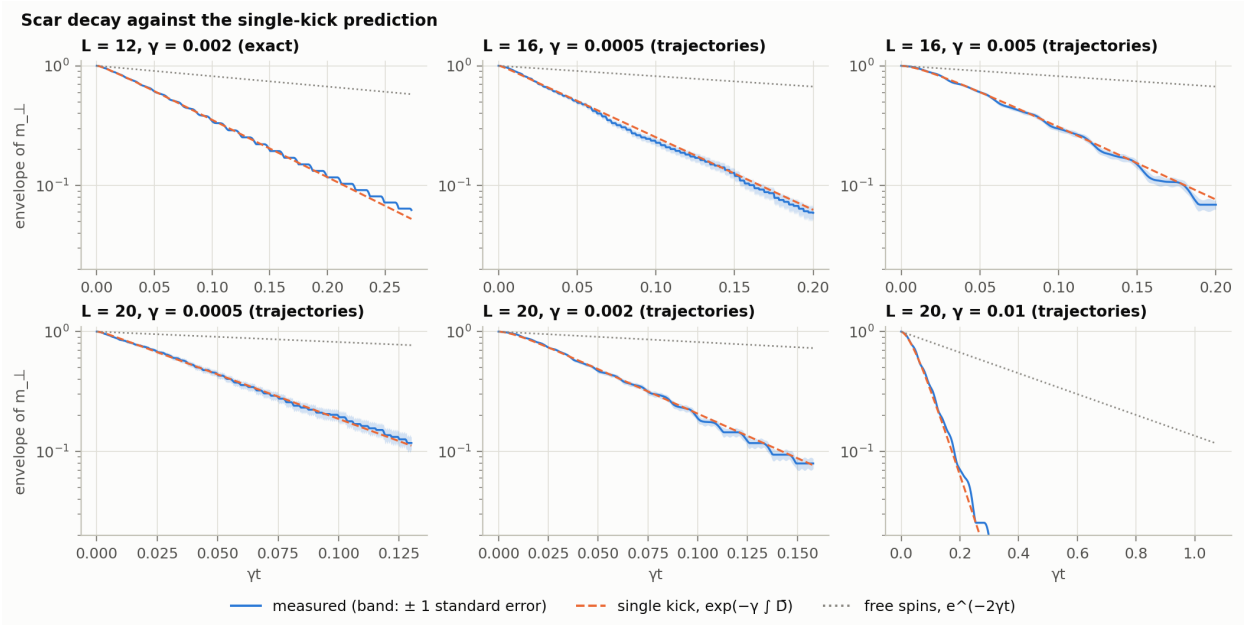


Figure 12: Envelope of $m_\perp$ (band: ± 1 standard error) against the single-kick prediction (49) and $e^{-2\gamma t}$ (`single_kick.py`).

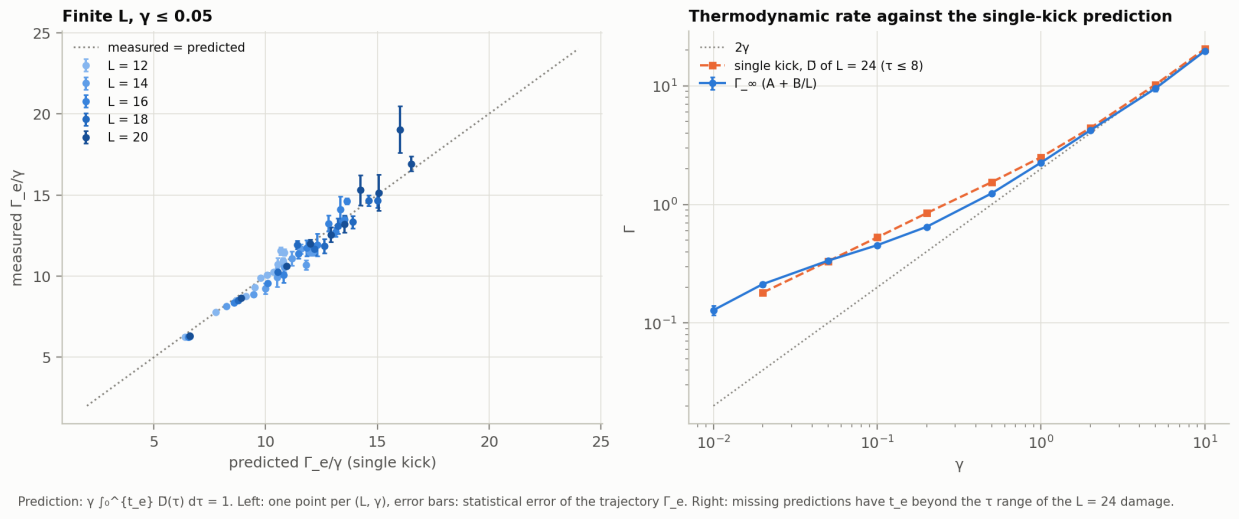


Figure 13: Left: measured against predicted Γ_e/γ at finite L. Right: Γ_∞ against the single-kick prediction from $\bar{D}$ at $L = 24$ (`single_kick.py`).

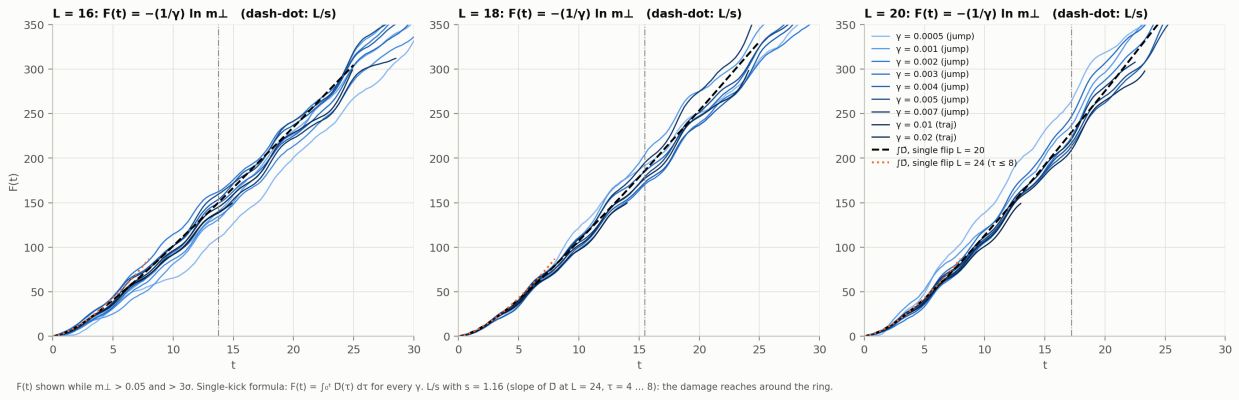


Figure 14: $F(t) = -\gamma^{-1} \ln m_\perp$ for every $\gamma \leq 0.02$ (jump trajectories for $\gamma < 0.01$, diffusive for 0.01, 0.02) at $L = 16, 18, 20$, shown while $m_\perp > 0.05$ and $> 3\sigma$; dashed: $\int_0^t \bar{D}$ of the same L; dotted: $\int_0^t \bar{D}$ of $L = 24$ ($\tau \leq 8$); dash-dot: $t = L/s$ (`plot_scripts/lean_plan_analysis.py`).

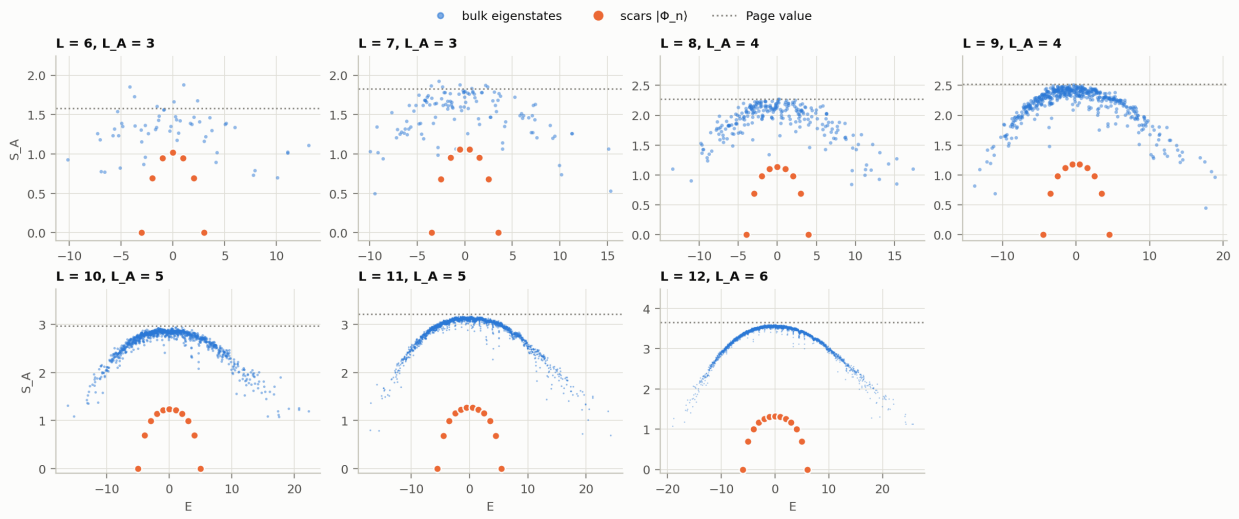


Figure 15: Half-chain entropy of all eigenstates of H against their energy, $L = 6$–12; scars $|\Phi_n\rangle$ in orange, Page value dotted (`entanglement`, `plot_entanglement.py`).

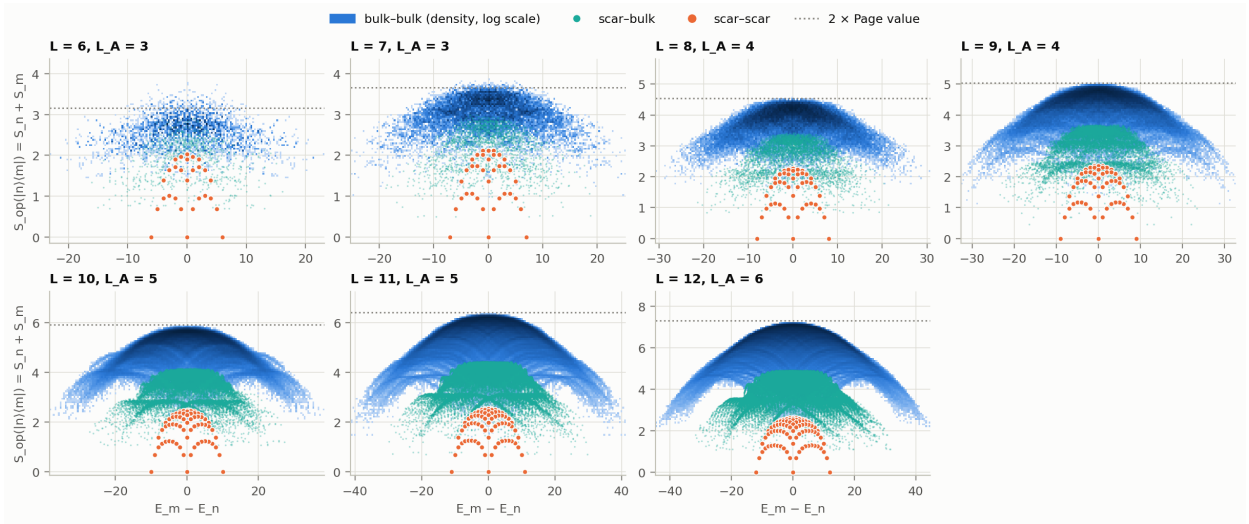


Figure 16: Operator entanglement $S_n + S_m$ of $|n\rangle\langle m|$ for all eigenstate pairs against $E_m - E_n$: bulk–bulk as a density, scar–bulk and scar–scar as points; twice the Page value dotted (`plot_operator_entanglement.py`).

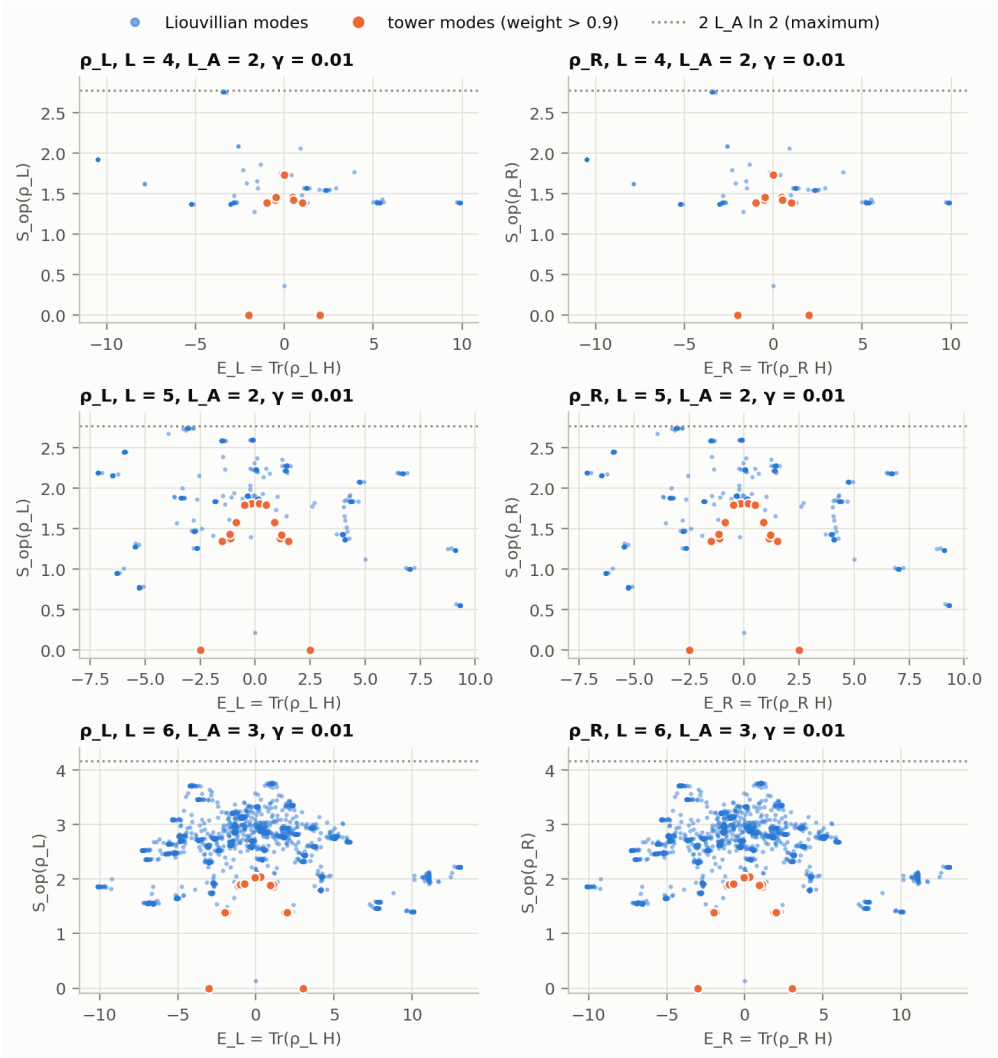


Figure 17: Operator entanglement of ρ_L (left) and ρ_R (right) of every Liouvillian mode against its energy, $\gamma = 0.01$, $L = 4, 5, 6$; tower weight > 0.9 in orange, $2L_A \ln 2$ dotted (`lindblad_entanglement`, `plot_lindblad_entanglement.py`; the $\gamma = 0$ figure, $L = 4$–7, is `lindblad_entanglement_g0.png`).

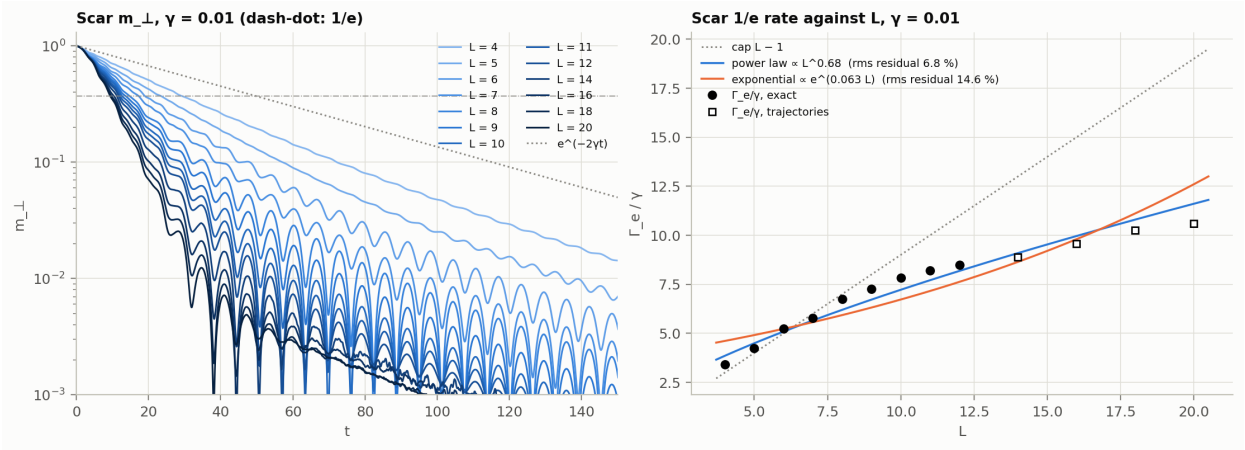


Figure 18: Current coefficients, random c, $\gamma = 0.01$: $m_\perp$ from $|-\rangle^{\otimes L}$ for $L = 4, \ldots, 20$ (exact up to 12, trajectories above) and Γ_e/γ against L: 3.40, 4.24, 5.24, 5.77, 6.73, 7.27, 7.84, 8.20, 8.49 ($L = 4, \ldots, 12$), 8.88, 9.56, 10.24, 10.61 ($L = 14, \ldots, 20$). A power law fits better ($\propto L^{0.68}$, rms residual 6.8%) than an exponential (14.6%); the points follow the cap $L - 1$ up to $L \approx 8$ and fall below it after (`plot_scripts/plot_rates_vs_L.py`). With the previous coefficients the window fits gave $\propto L^{1.13}$ for $L \leq 10$ (Table 13).

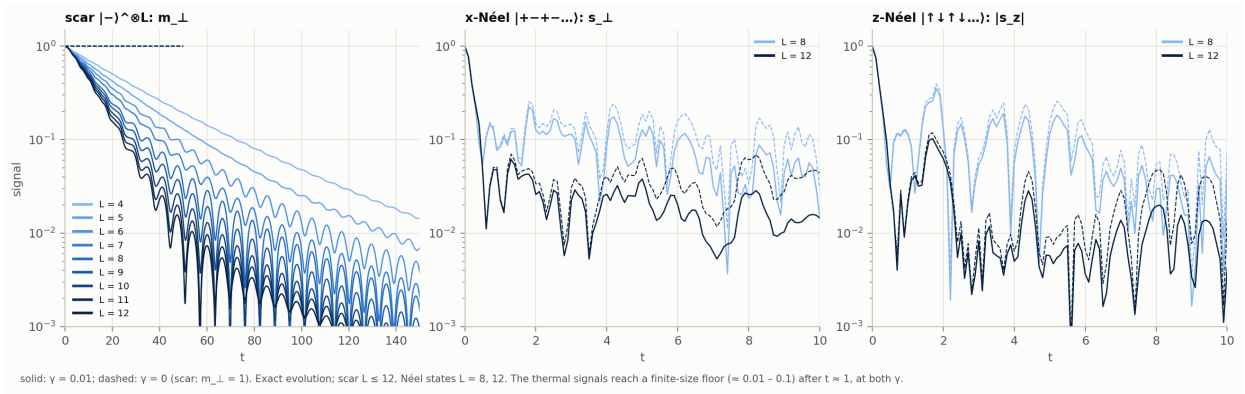


Figure 19: Current coefficients: the scar $m_\perp$ (P1) and the two thermal references, x-Néel $s_\perp$ (in place of P3) and z-Néel $|s_z|$, against time, $\gamma = 0.01$ (solid) and $\gamma = 0$ (dashed); exact evolution (`plot_rates_vs_L.py`). The thermal signals drop within $t \approx 1$ to a finite-size floor, at both γ; the scar decays only through the dephasing.

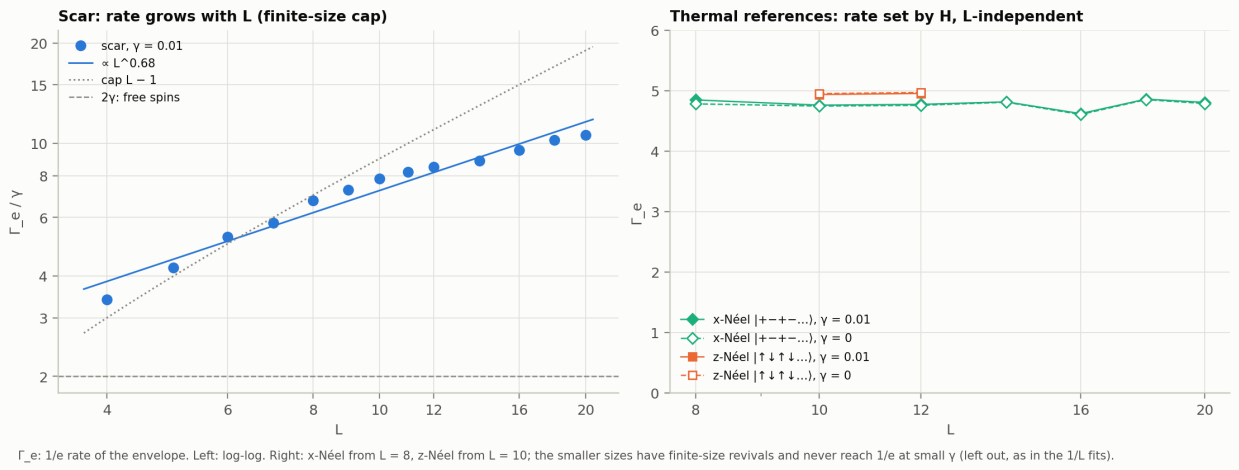


Figure 20: Current coefficients: Γ_e against L. Left: scar Γ_e/γ at $\gamma = 0.01$ (log-log) with the power law, the cap $L-1$ and the free-spin value 2. Right: x-Néel ($L \geq 8$) and z-Néel ($L \geq 10$) Γ_e at $\gamma = 0$ and 0.01: 4.6–5.0, independent of L and of γ, set by the closed-system thermalization (`plot_rates_vs_L.py`).

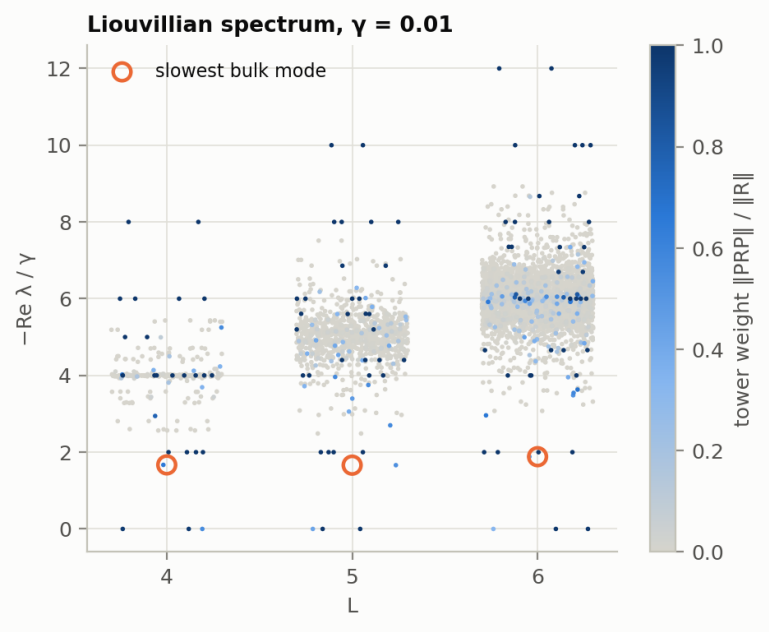


Figure 21: $-\mathrm{Re}\,\lambda/\gamma$ of all Liouvillian modes, coloured by tower weight; circle: slowest bulk mode (`plot_scripts/plot_spectrum.py`).